Outsourcing Extinction: Economic Complexity and the Geography of Biodiversity Loss*

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Abstract

Does the structure of an economy, beyond its level of income, shape biodiversity loss? We relate the Red List Index of aggregate species extinction risk to economic complexity for up to 179 countries between 1993 and 2019. Conditional on income per capita, greater economic complexity raises extinction risk within countries: a one-standard-deviation increase in the Economic Complexity Index lowers the Red List Index by about one to two percent. The result is robust to country and period fixed effects, a long-difference specification, sectoral controls, and instrumental-variables estimation using product-space diversification potential. The mechanism is not domestic. Agricultural land use, forest cover, input intensity, and agricultural productivity do not mediate the relationship, nor are they systematically affected by economic complexity. Instead, complex economies displace biodiversity pressure abroad. In a bilateral matrix of extinction-risk footprints, they are net importers of biodiversity loss, sourcing an increasing share of their footprint from other countries' species. Imports from biome-sharing regions predict subsequent declines in domestic extinction-risk indices, evidence for a range-overlap channel linking foreign habitat conversion to home-country biodiversity outcomes. Economic complexity therefore reorganizes the geography of extinction risk rather than reducing it. Structural transformation does not, on its own, deliver decoupling from biodiversity loss, and conservation policies based on territorial indicators understate the biodiversity cost of complex economies.

Keywords: Biodiversity; Extinction risk; Red List Index; Economic complexity; Structural transformation; Trade-embodied footprint

JEL classification: Q56; Q57; O11; O14; F18

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1 Introduction

Species are disappearing tens to hundreds of times faster than the average of the past ten million years, and economic activity is the proximate cause (IPBES 2019, Díaz et al. 2019, Pimm et al. 2014). The principal pressures behind biodiversity loss (land- and sea-use change, direct exploitation, climate change, pollution, and invasive species) all arise from how economies produce and consume (IPBES 2019, Maxwell et al. 2016). Yet the literature has framed this question almost entirely in terms of income: a large body of work tests for an Environmental Kuznets Curve (EKC) in biodiversity, an inverted-U in which threat first rises and then falls as countries grow richer (Naidoo & Adamowicz 2001, Asafu-Adjaye 2003, Dietz & Adger 2003, Mills & Waite 2009). This perspective focuses on the scale of economic activity. It says considerably less about its composition. Economies with similar income can differ markedly in the goods they produce and the knowledge embedded in their productive systems, and those differences may shape biodiversity pressure as much as income itself. This paper therefore asks whether economic complexity—the diversity and sophistication of a country's productive structure—shapes biodiversity threat, holding income constant.

Economic complexity provides a natural framework for this question because it captures the productive capabilities embodied in what countries make and export (Hidalgo & Hausmann 2009, Hausmann et al. 2007, Hidalgo 2021). Beyond explaining differences in growth and inequality (Hausmann et al. 2007, Hartmann et al. 2017), productive structure may also shape how economic activity bears on natural ecosystems. As economies become more complex, production typically shifts away from a narrow base of resource- and land-intensive activities toward a broader portfolio of knowledge-intensive industries. This transformation could, in principle, ease pressure on nature, consistent with evidence that complexity lowers carbon emissions and supports cleaner production in advanced economies (Romero & Gramkow 2021, Mealy & Teytelboym 2022, Stojkoski et al. 2023). We find the opposite: holding income constant, greater complexity is associated with higher biodiversity threat, suggesting that what economies produce can matter as much as how much they produce.

Our first result is that economic complexity raises biodiversity threat, conditional on income. Relating the Red List Index of aggregate extinction risk (Butchart et al. 2004, 2007) to the ECI for up to 179 countries between 1993 and 2019, we find that more complex economies face higher extinction risk among their own species. The estimate is robust to country and period fixed effects, to a long-difference design that removes all time-invariant national heterogeneity, to controlling for the full sectoral composition of output, to the temporal aggregation of the data, and to the choice of complexity measure. Reverse causation is unlikely, because export structure is predetermined: it is built from capabilities accumulated over decades, not from the current status of a country's species. Instrumenting complexity with its product-space diversification potential leaves a negative effect of similar or larger magnitude on a strong first stage. Sophistication, on this evidence, does not appear to spare a country's own biodiversity.

Our second result is that this effect does not run through the obvious domestic channel. If complex economies threatened their own species, one would expect the effect to operate through their own land use: more cropland, less forest, more intensive fertilizer and pesticide use, higher agricultural productivity, or faster deforestation. None of these domestic channels accounts for the complexity effect. Across every specification the complexity coefficient is unchanged, the channels are not themselves moved by complexity, and none mediates the relationship. The pressure that complexity exerts on biodiversity is not visible in a country's own land. This does not make land use irrelevant; it suggests that the land being converted lies abroad.

Our third result locates that pressure abroad. A growing literature traces the species threat and extinction risk embodied in international supply chains, showing that consumption in one country drives biodiversity loss in others (Lenzen et al. 2012, Moran & Kanemoto 2017, Marques et al. 2019, Wilting et al. 2017, Chaudhary & Kastner 2016). Using a bilateral matrix of extinction-risk footprints (Irwin et al. 2022), we find that more complex economies are net importers of extinction risk: they source a growing share of their biodiversity footprint from other countries' species. Economies in the top tercile of complexity place about ninety percent of their consumption footprint abroad and are net importers of extinction risk, while the least complex economies place far less abroad and are net exporters. Complexity, on this reading, does not reduce the biodiversity cost of consumption so much as relocate it to the ecosystems of trading partners. Familiar commodities make the point concrete: oil palm, soy, and the metals of the energy transition each convert biodiverse habitat far from where the final good is consumed, yet none of these costs appears in the importing country's domestic land accounts.

Taken together, these three results form a single contribution: structural transformation relocates extinction risk rather than reducing it. This reframes a debate that has centered on income. The biodiversity-EKC literature, by relying on cross-sectional income variation, compares the experience of poor and rich countries, in which the relocation of damage across borders can be hard to distinguish from genuine abatement (Stern 2004, Vincent 1997). The complexity-and-environment literature has established that sophistication can lower territorial emissions (Romero & Gramkow 2021) even as it raises aggregate ecological pressure (Neagu 2020), and the corresponding question for biodiversity in particular remains open. We find that it does not, because biodiversity loss travels through trade in a way that carbon accounting within borders does not capture (Lenzen et al. 2012, Wiebe & Wilcove 2025, Li & Chen 2024). The policy implication follows directly: conservation targets and disclosure frameworks anchored in domestic indicators will tend to understate the biodiversity cost of the world's most complex economies.

The framework that ties these results together brings the trade-and-environment model of Copeland & Taylor (1994, 2004) into contact with the species–area relationship from ecology (Preston 1962, Pimm & Raven 2000, Brooks et al. 2002) and economic complexity as a source of comparative advantage and productivity (Hidalgo & Hausmann 2009, Hausmann et al. 2007). The key link is the species-range overlap matrix, ω_{ij}, which measures the extent to which the species occurring in country i also depend on habitat in country j. Through this ecological bridge, habitat conversion embodied in trade propagates extinction risk across borders whenever species ranges span multiple countries. The framework delivers a biodiversity index constructed directly from the conservation status of the species present in each country and yields four testable predictions: economic complexity raises a country's own extinction risk; domestic land-use controls do not mediate this relationship; more complex economies become net importers of extinction risk; and biodiversity outcomes exhibit positive spatial dependence. None of these predictions emerges from trade theory, species–area relationships, or economic complexity in isolation; they arise only by integrating the three into a unified framework.

Related Literature. The framework naturally connects three literatures that have evolved largely in parallel: the empirical literature on economic growth and biodiversity loss, the economics of productive complexity, and the trade-embodied accounting of biodiversity footprints. This paper contributes to each. The first asks whether growth threatens or spares biodiversity, and frames the question in terms of the level of income. Borrowing the Environmental Kuznets Curve logic developed for pollution, a large empirical literature looks for an inverted-U between income and species threat, with threat rising among poor countries and falling once they are rich enough (Naidoo & Adamowicz 2001, Asafu-Adjaye 2003, Dietz & Adger 2003, Mills & Waite 2009). The evidence is mixed, and it rests largely on cross-country comparisons of poor and rich nations at a point in time. Stern (2004) noted for carbon that such cross-sectional curves can in part reflect confounding or the relocation of damage rather than its abatement, a caution that may apply with particular force to biodiversity, whose damage is fixed to the converted habitat. Our approach differs from this strand in both what we ask and how we identify it. We hold income constant and vary the structure of production, and we estimate the relationship within countries over time, where complexity raises extinction risk even though the cross-section is flat.

The second strand is the economics of complexity. Building on the product space and the idea that what a country exports reveals the capabilities it has accumulated, this literature measures economic complexity from the diversity and ubiquity of export baskets (Hidalgo & Hausmann 2009, Hausmann et al. 2007, Hidalgo 2021) and shows that it predicts future growth and inequality beyond what factor accumulation explains (Hartmann et al. 2017, Mealy & Teytelboym 2022). A more recent branch turns to the environment and finds, for carbon, that more complex economies emit less per unit of output and innovate toward cleaner technologies, so that sophistication and territorial decarbonization tend to move together (Romero & Gramkow 2021, Boleti et al. 2021). We carry this literature to biodiversity and find that the environmental dividend does not appear to generalize: complexity raises extinction risk. The contrast with carbon need not be a puzzle, because biodiversity damage, unlike emissions, is local to the habitat and is displaced through trade rather than abated.

The third strand quantifies the biodiversity loss embodied in international trade. Using multi-regional input–output and bilateral footprint accounts, it shows that consumption in wealthy economies drives a large and rising share of the species threat recorded in their trading partners, concentrated in the biodiverse tropics (Lenzen et al. 2012, Moran & Kanemoto 2017, Marques et al. 2019, Wilting et al. 2017, Chaudhary & Kastner 2016, Irwin et al. 2022, Wiebe & Wilcove 2025). A complementary line of work uses sub-national variation within countries to relate local economic activity to species imperilment (Liang et al. 2021). Both literatures establish the geography of the problem and are largely descriptive in focus, mapping where consumption-driven biodiversity loss falls. We build on it by adding two further questions: which structural feature of an economy makes it a large net exporter of biodiversity pressure, and whether the footprint a country imports is reflected in the extinction risk it records at home. We show that economic complexity is a systematic structural driver of how much biodiversity pressure a country displaces abroad, and that the most externalized economies tend to be those whose own species grow more threatened. A trade model in which complexity shifts comparative advantage toward land-saving goods, joined to a species–area map with cross-border ranges, draws these facts together and frames the policy question of consumption-based accountability (Vergez et al. 2025, Dasgupta 2021).

Outline. The remainder of the paper is organized as follows. Section 2 develops the theoretical framework and derives its testable predictions. Section 3 describes the data and empirical strategy. Section 4 presents the empirical evidence, examining the relationship between economic complexity and biodiversity threat, the role of trade-embodied extinction risk, and a range of robustness and mechanism analyses. Section 5 discusses the implications of the findings for the biodiversity-growth literature and conservation policy. Section 6 concludes.

2 Theoretical framework

This section develops a two-region trade model in the tradition of Copeland & Taylor (1994, 2004), augmented with a species–area relationship linking habitat conversion to extinction risk (Pimm & Raven 2000, Brooks et al. 2002). The model generates the empirical results of the paper as comparative-statics predictions. It further characterizes the inefficiency of the decentralized equilibrium and derives the policy instrument that restores efficiency.

2.1 Environment

Two regions, Home (H) and Foreign (F), trade two final goods. Good x is habitat-intensive: producing it converts natural land that would otherwise shelter species (agriculture, pasture, and resource extraction). Good y is knowledge-intensive and land-saving (sophisticated manufactures and services). Each region i is endowed with labor L_i, mobile across its sectors, and a stock of convertible habitat $\bar{A}_i$.

Preferences. Preferences are identical and homothetic. A representative consumer in region i values the two goods and the biodiversity present in its territory,

$$U_i = \beta \log c_i^x + (1-\beta)\log c_i^y + \phi\, b_i, \qquad \beta \in (0,1),\ \phi \geq 0, \tag{2.1}$$

where c_i^x, c_i^y are consumption levels, b_i is the conservation status of the species present in i (defined in Section 2.2), and ϕ is the weight placed on it. Cobb–Douglas preferences over goods fix expenditure shares, so that $p_x c_i^x = \beta I_i$ and $p_y c_i^y = (1-\beta)I_i$, where I_i is income and (p_x, p_y) are world prices. This is the property that lets the model speak to regressions that condition on income: at a given income and given prices, consumption is pinned down, so complexity can act only through the composition and the location of production.

Technology. Technology uses labor under diminishing returns, with each sector employing a fixed complementary factor (knowledge capital in y, land in x), following a specific-factors structure:

$$y_i = \kappa_i\,(L_i^y)^\eta, \qquad x_i = B\,(L_i^x)^\eta, \qquad L_i^x + L_i^y = L_i, \qquad \eta \in (0,1). \tag{2.2}$$

Producing the habitat-intensive good converts land in fixed proportion, $h_i = \gamma_i x_i$, where h_i is hectares of habitat lost and $\gamma_i > 0$ its land intensity. We interpret h_i broadly. Cropland cleared for export agriculture is the most direct case, but the same logic covers the forest opened by logging, the ground broken by mining for the metals of the green transition, the rivers and soils contaminated by industrial and agricultural runoff, and the roads and ports laid through ecosystems to move what these activities produce. The species–area map below treats them as equivalent because each subtracts effective habitat from the species of region i. The parameter κ_i is economic complexity: the diversity and sophistication of a region's capabilities, which raise productivity in the knowledge-intensive sector. This is the reduced-form image of the product space of Hidalgo & Hausmann (2009), Hausmann et al. (2007), in which complex economies command the capabilities to produce many sophisticated, land-saving goods. Higher κ_i raises the relative return to allocating labor to y and therefore the relative cost of producing x, giving complex economies a comparative advantage in land-saving goods.

2.2 Habitat, species, and conservation status

Converted habitat drives extinction through the species–area relationship, one of the most robust regularities in ecology (Pimm & Raven 2000, Brooks et al. 2002). With original habitat $\bar{A}_i$ and conversion h_i, the surviving habitat is $A_i = \bar{A}_i - h_i$, and the share of species in region i committed to extinction is

$$E_i = 1 - \left(\frac{A_i}{\bar{A}_i}\right)^z = 1 - \left(1 - \frac{h_i}{\bar{A}_i}\right)^z, \qquad z \in (0,1), \tag{2.3}$$

increasing and convex in conversion, with the canonical exponent $z \approx 1/4$ (derived theoretically by Preston 1962, Brooks et al. 2002, Pimm & Raven 2000, and routinely used in habitat-loss extinction predictions).

Species ranges cross borders, so the biodiversity status of species present in one region depends on habitat loss in others. Let $\Omega = (\omega_{ij})_{i,j}$ denote the species-range overlap matrix, where each element $\omega_{ij} \geq 0$ is the share of region i's species assemblage whose ranges include region j. Rows sum to one, $\sum_j \omega_{ij} = 1$. The term ω_{ii} captures locally endemic species, while ω_{ij} for $j \neq i$ captures cross-border overlap arising from shared, migratory, or biome-spanning ranges. The biodiversity index of region i is the overlap-weighted share of its species not yet committed to extinction,

$$b_i = 1 - \sum_j \omega_{ij} E_j = \sum_j \omega_{ij} \left(1 - \frac{h_j}{\bar{A}_j}\right)^z. \tag{2.4}$$

The marginal effect of conversion in region j on the biodiversity index of region i is

$$\frac{\partial b_i}{\partial h_j} = -\,\omega_{ij} m_j, \qquad m_j \equiv \frac{z}{\bar{A}_j}\left(1 - \frac{h_j}{\bar{A}_j}\right)^{z-1} > 0, \tag{2.5}$$

where m_j is the marginal extinction intensity of region j: the species lost per hectare converted. It is larger where habitat is biodiverse and intact, so conversion relocated into species-rich frontiers is especially damaging. The empirical footprint accounts, in which the consumption of rich economies falls on biodiverse tropical habitat (Lenzen et al. 2012, Chaudhary & Kastner 2016, Irwin et al. 2022), motivate the following ranking.

Definition 1 (Empirically motivated ranking). *Home is the complex, already-converted region; Foreign is the biodiverse, less-converted one. Two conditions hold at the relevant margin: (i) Foreign's marginal extinction intensity is at least as large as Home's, $m_F \geq m_H$, so each additional hectare converted in Foreign costs at least as many species as one in Home; and (ii) species ranges overlap across borders, $\omega_{HF} > 0$.*

2.3 Trade equilibrium and the composition of production

Take world prices as given and normalize $p_y = 1$, writing $p \equiv p_x$ for the relative price of the habitat-intensive good. Cost minimization equates the value marginal product of labor across sectors, $\kappa_i \eta (L_i^y)^{\eta-1} =$

$pB\eta(L_i^x)^{\eta-1}$, so that

$$\frac{L_i^y}{L_i^x} = \left(\frac{\kappa_i}{pB}\right)^{\frac{1}{1-\eta}}, \qquad L_i^x = \frac{L_i}{1+(\kappa_i/pB)^{1/(1-\eta)}}. \tag{2.6}$$

Labor reallocates toward the knowledge-intensive sector as complexity rises, which establishes the central supply-side property of the model.

Lemma 2.1 (Composition effect)**.** *At given prices, the supply of the habitat-intensive good and the habitat converted to produce it are strictly decreasing in complexity: $\partial x_i/\partial\kappa_i < 0$ and $\partial h_i/\partial\kappa_i = \gamma_i\,\partial x_i/\partial\kappa_i < 0$.*

The proof is immediate from Equation (2.6), since L_i^x falls in κ_i and $x_i = B(L_i^x)^\eta$, $h_i = \gamma_i x_i$; the explicit derivative is in Appendix A.1.

The two regions trade to clear world markets. Home is the complex region, $\kappa_H > \kappa_F$, and by Lemma 2.1 it produces relatively more y and less x. World market clearing for the habitat-intensive good,

$$x_H(p,\kappa_H) + x_F(p,\kappa_F) = c_H^x + c_F^x, \tag{2.7}$$

together with balanced trade determines the equilibrium price p. A rise in κ_H contracts Home's supply of x, raises its relative supply of y, and so raises the equilibrium relative price p of the habitat-intensive good; Foreign expands its production of x in response, and Home meets a larger part of its fixed consumption c_H^x through imports. Habitat conversion therefore migrates from Home to Foreign as Home grows more complex: $dh_H/d\kappa_H < 0$ and $dh_F/d\kappa_H > 0$ (see Appendix A.3).

Define the conditional effect of complexity as the change in Home's biodiversity index holding income I_H and prices constant,

$$\left.\frac{d\,b_H}{d\kappa_H}\right|_{I_H} = -\,\omega_{HH}\,m_H\,\frac{dh_H}{d\kappa_H} \;-\; \omega_{HF}\,m_F\,\frac{dh_F}{d\kappa_H}, \tag{2.8}$$

obtained by differentiating Equation (2.4) and using Equation (2.5) (see Appendix A.2). The first term captures the domestic channel, and the second the foreign channel. Holding income I_H fixed isolates the structural effect of complexity, which operates through the composition of production and the location of habitat conversion. This conditional derivative is the theoretical counterpart of the regression coefficient on the Economic Complexity Index, conditional on income.

2.4 Complexity and extinction risk

Equation (2.8) carries the central argument. The two terms are forces of opposite sign. The first is positive: complexity lowers domestic conversion, which on its own would push Home's biodiversity index up. The second is negative: complexity raises foreign conversion, which through shared ranges pushes the index down. The sign of the total depends on which force dominates, and Definition 1 settles the question.

Proposition I (Complexity raises own extinction risk)**.** *Holding income and prices fixed,* $d\,b_H/d\,\kappa_H|_{I_H} <$ 0 *whenever*

$$\omega_{HF}\, m_F\, \frac{dh_F}{d\kappa_H} \;>\; \omega_{HH}\, m_H \left(-\frac{dh_H}{d\kappa_H}\right). \tag{2.9}$$

Under Definition 1 this holds, so a more complex economy faces higher extinction risk among its own species.

The intuition is that a complex economy cannot fully insulate its species from its own consumption. Reducing domestic conversion spares only locally endemic species, weighted by $\omega_{HH}\, m_H$, while the displaced conversion shifts to biodiverse foreign habitat, weighted by $\omega_{HF}\, m_F$, and feeds back to Home through overlapping ranges. When this displaced damage is sufficiently large, because the frontier is species-rich (m_F is high) and ranges are widely shared ($\omega_{HF} > 0$), Home's biodiversity index declines. This corresponds to the within-country, income-conditional relationship estimated in Section 4.

Corollary 2.1 (The domestic channel does not mediate)**.** *By Lemma 2.1,* $dh_H/d\kappa_H \leq 0$*, so the domestic term* $-\omega_{HH}\, m_H\, (dh_H/d\kappa_H)$ *in Equation (2.8) is non-negative. Conditioning the effect of complexity on domestic land use therefore leaves it negative; in the limit in which Home ceases to produce the habitat-intensive good, the domestic term vanishes and the effect runs entirely through the foreign channel,* $\partial\, b_H/\partial\, \kappa_H|_{h_H} = -\,\omega_{HF}\, m_F\, (dh_F/d\kappa_H) < 0$*.*

Because complexity lowers rather than raises domestic conversion, no domestic land-use variable can account for the effect, and controlling for such variables cannot displace it. This is the empirical prediction tested in Section 4.3: agricultural and land-cover controls leave the complexity coefficient intact, because the model implies that the mediating channel is foreign, not territorial.

Proposition II (Leakage and net imports of extinction risk)**.** *Let* s_H *be the share of the extinction risk embodied in Home's consumption that is incurred abroad, and* NIE_H *its net imported extinction risk (the risk its consumption causes abroad minus the risk foreign consumption causes within it). Then* $ds_H/d\kappa_H > 0$ *and* $d\mathrm{NIE}_H/d\kappa_H > 0$*.*

As Home grows more complex it produces less of the habitat-intensive good and imports more, so a rising share of the habitat conversion behind its consumption occurs in Foreign, and its net position turns toward importing extinction risk. This is the cross-sectional gradient documented in Section 4.4.

Corollary 2.2 (Spatial dependence)**.** *The cross-country biodiversity indices* $\{b_i\}_{i=1}^N$ *are positively spatially correlated whenever range overlap is non-trivial. Treating* $(1-h_j/\bar{A}_j)^z$ *as random across countries with finite variance,*

$$\mathrm{Cov}(b_i, b_k) \;=\; \sum_{j,\ell} \omega_{ij}\, \omega_{k\ell}\, \mathrm{Cov}\big((1-h_j/\bar{A}_j)^z,\, (1-h_\ell/\bar{A}_\ell)^z\big) \;\geq\; 0, \tag{2.10}$$

with strict inequality whenever there exists j *such that* $\omega_{ij}\, \omega_{kj} > 0$*. The best linear projection of* b_i *onto the spatial lag* $\sum_{k\neq i} W_{ik}\, b_k$ *then takes the spatial-autoregressive form*

$$b_i \;=\; \rho \sum_{k\neq i} W_{ik}\, b_k \;+\; \xi_i, \tag{2.11}$$

where W is a row-standardized weight matrix derived from Ω, $\rho \geq 0$ is the projection coefficient, and ξ_i is the residual orthogonal to the spatial lag. Appendix A.4 gives the derivation.

Shared ranges make a country's biodiversity status partly a function of its neighbors'. When habitat is converted in one country it commits species to extinction wherever their range extends, so countries that share biogeographic space share biodiversity outcomes. The corollary states this formally: biodiversity indices satisfy a spatial-autoregressive relation in which each country's index loads positively on the indices of the regions it overlaps with.

2.5 Welfare and policy

The two-region setting introduced above is sufficient to derive the within-country and leakage results, because the relevant economic forces are essentially dyadic: Home's complexity reallocates land-intensive production between Home and the rest of the world. The welfare analysis, however, requires the full multilateral structure. The externality borne by region i from habitat conversion in region j runs through the bilateral overlap weight ω_{ij}, and the social marginal cost of conversion in j aggregates these damages across all affected regions. With only two regions, the cross-border externality collapses to a single bilateral term and the policy implications would understate the global scope of the wedge. We therefore generalize the model to N regions indexed by $i = 1, \ldots, N$, each with population $n_i > 0$ that supplies the labor endowment L_i used in production. All other model objects (technologies, preferences, and the species-range overlap matrix Ω) carry over without modification. The empirical bilateral footprint matrix used in Section 4.5 covers 161 countries, providing an empirical counterpart to this N-region structure.

The decentralized equilibrium is inefficient because extinction externalities generated by consumption are not priced. To make this precise, consider a planner who chooses the world allocation $\{c_i^x, c_i^y, L_i^x, L_i^y\}_{i=1}^N$ to maximize world welfare

$$\mathscr{W} = \sum_i n_i \left[\beta \log c_i^x + (1-\beta) \log c_i^y + \phi\, b_i\right], \tag{2.12}$$

subject to the technology in Equation (2.2), the habitat-conversion identity $h_i = \gamma_i x_i$, the biodiversity index in Equation (2.4), and goods-market clearing $\sum_i c_i^x = \sum_i x_i$ and $\sum_i c_i^y = \sum_i y_i$.

The planner's first-order condition with respect to h_j yields the social marginal cost of habitat conversion in region j,

$$\mathrm{SMC}_j \equiv -\sum_i n_i \phi \frac{\partial b_i}{\partial h_j} = \phi\, m_j \sum_i n_i\, \omega_{ij}, \tag{2.13}$$

which aggregates all populations exposed to species loss originating in j through range overlap. In the decentralized equilibrium, this global exposure is not internalized: agents in region j price only the local component of the damage transmitted to their own biodiversity.

A consumer in region ℓ whose demand induces conversion in j internalizes only the privately borne component

$$\mathrm{PMC}_{\ell j} = \phi\, n_\ell\, \omega_{\ell j}\, m_j. \tag{2.14}$$

The resulting wedge,

$$\Delta_{\ell j} \equiv \mathrm{SMC}_j - \mathrm{PMC}_{\ell j} = \phi\, m_j \sum_{i \neq \ell} n_i\, \omega_{ij} \geq 0, \tag{2.15}$$

captures uninternalized third-party exposure: extinction risk generated in j is transmitted to all other regions with overlapping species ranges but is not reflected in private decision-making.

Proposition III (Inefficiency and consumption-based correction)**.** *The decentralized equilibrium is inefficient whenever $\Delta_{\ell j} > 0$ for some pair (ℓ, j). A territorial Pigouvian tax charges each region only for damage to its own species,*

$$t_j = \phi\, n_j\, \omega_{jj}\, m_j,$$

thereby pricing only the local component of SMC_j *and leaving unpriced the cross-border externality $\sum_{i \neq j} n_i\, \omega_{ij}\, m_j$. This policy corrects only the diagonal of the externality matrix.*

The first-best allocation is implemented by a consumption-based charge

$$\tau_j = \mathrm{SMC}_j = \phi \sum_i n_i\, \omega_{ij}\, m_j, \tag{2.16}$$

levied on each unit of habitat conversion in j induced by consumption, including imported demand. Under τ, private first-order conditions coincide with the planner's, and the allocation is efficient. An equivalent decentralization is a set of transfers from importing to exporting regions equal to $\Delta_{\ell j}$ for each pair (ℓ, j).

The policy interpretation aligns with the wealth- and footprint-based accounting emphasized by Dasgupta (2021): conservation targets and disclosure frameworks based on domestic indicators internalize only $\phi\, n_j\, \omega_{jj}\, m_j$ and therefore understate the full cost of production in complex economies, where a large share of biodiversity loss is transmitted through off-diagonal terms $\phi\, n_i\, \omega_{ij}\, m_j$ for $i \neq j$. Appendix A.5 provides the formal proof via the planner's Lagrangian.

3 Data and Empirical Strategy

The theoretical framework yields four main predictions: economic complexity reduces the biodiversity index within countries conditional on income (Proposition I); this effect is not mediated by domestic land-use controls (Corollary 2.1); complex economies are net importers of extinction risk (Proposition II); and biodiversity indices exhibit positive spatial dependence (Corollary 2.2). Proposition III then provides the welfare interpretation of these mechanisms.

The empirical analysis is structured around these implications. We proceed in three steps. First, we estimate the within-country relationship between economic complexity and extinction risk, conditional on income. Second, we examine whether domestic land-use adjustments account for this relationship. Third, we test whether economic complexity is associated with the displacement of biodiversity pressure across borders through trade.

3.1 Data

The dependent variable is the Red List Index (RLI), which tracks aggregate extinction risk across groups of species (Butchart et al. 2004, 2007, Hoffmann et al. 2010). The index ranges from zero, when all species in a group are extinct, to one, when all are of Least Concern; higher values therefore indicate lower extinction risk. It is the empirical counterpart of the conservation-status index b_i in the theoretical framework and underpins SDG indicator 15.5.1. By construction, it changes only when species are reassigned to different IUCN Red List categories following formal reassessments (IUCN 2012, BirdLife International & Handbook of the Birds of the World 2019).

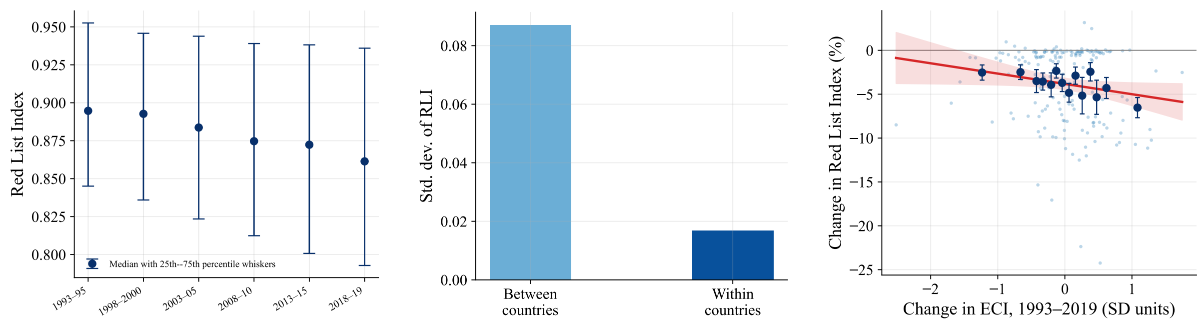
Two features of the RLI are central for interpretation. First, it is defined at the level of global species status: a species assigned a given risk category contributes that same status to every country in which it is present. Second, species ranges are highly transboundary: major biomes such as the Amazon, Congo, and Southeast Asian rainforests span multiple countries, and many taxa are migratory or widely distributed. As a result, a country's RLI reflects habitat loss wherever its species occur, not only within its borders. In the sample, the RLI averages 0.87, ranging from 0.42 (Mauritius) to near unity (Finland and Sweden).

The main explanatory variable is the Economic Complexity Index (ECI), constructed by the Growth Lab at Harvard University from countries' export structures (Hidalgo & Hausmann 2009, Hausmann et al. 2007, 2014, Growth Lab at Harvard University 2024). The index is high for economies exporting diverse and sophisticated products that are rare globally, and low for economies concentrated in a narrow set of ubiquitous, typically resource- or land-intensive goods. We standardize the ECI so that coefficients correspond to a one-standard-deviation change. Because it reflects accumulated productive capabilities embedded in export structure, it evolves slowly over time and is less likely than contemporaneous income to be affected by short-run feedback from extinction dynamics (Hidalgo et al. 2007, Hidalgo 2021).

Control variables include GDP per capita, population density, and protected-area coverage from the World Bank (The World Bank 2020), and the Human Development Index from UNDP (United Nations Development Programme 2020). The dataset is organized into six five-year periods between 1993 and 2019 (1993–95, 1998–2000, 2003–05, 2008–10, 2013–15, and 2018–19), constructed as two- or three-year averages to reduce noise in economic series. This temporal aggregation is motivated by the biology of the dependent variable: IUCN species are typically reassessed at frequencies of five to ten years (IUCN 2012, Butchart et al. 2007), implying limited meaningful annual variation in the RLI. Five-year spacing therefore represents the finest empirically meaningful resolution, and we show in Section 4.3 that results are robust to coarser aggregation. The panel covers up to 179 countries; a balanced six-period panel is used for robustness.

Figure 1 summarizes the empirical structure. Panel (a) shows that the RLI is high and relatively compressed across countries. Panel (b) documents that within-country variation is substantially smaller than cross-country variation, reflecting the slow reassessment cycle of extinction-risk categories. This motivates a fixed-effects and long-difference strategy, which identifies relationships primarily from medium-run changes. Panel (c) plots changes in RLI against changes in economic complexity over 1993–2019; the slope is negative, indicating that countries experiencing increases in complexity also experienced increases in extinction risk.

Mechanism variables include land-use shares (agricultural, forest, and urban), agricultural land extent, fertilizer and pesticide use, agricultural productivity and output, and forest loss indicators, drawn



(a) Median RLI by period **(b)** Between vs. within variation of RLI **(c)** ΔECI vs. ΔRLI, 1993–2019

Figure 1: Overview of the estimation sample.

Notes: Panel (a) plots the median Red List Index across countries in each of the six periods (1993–95, 1998–2000, 2003–05, 2008–10, 2013–15, 2018–19); the whiskers at each period span the 25th to the 75th percentile. Panel (b) decomposes the standard deviation of the Red List Index into its between-country and within-country components. Panel (c) plots, for each country, the change in the Red List Index against the change in economic complexity over 1993–2019 (horizontal axis in standard-deviation units; vertical axis the corresponding percentage change in the Red List Index). Each faint point is a country; solid markers are means within equal-count bins of complexity, with ± 1 standard-error bars, and the shaded band is the 95% confidence interval of the fitted line.

from the World Bank, FAO, and the ESA Climate Change Initiative (CCI) Land Cover dataset. For the leakage analysis, we use a bilateral extinction-risk footprint matrix for 2013, which allocates biodiversity pressure embodied in consumption across country pairs (Irwin et al. 2022, Lenzen et al. 2012, Moran & Kanemoto 2017). From this matrix, we construct domestic, imported, exported, and net extinction-risk footprints for each country.

3.2 Econometric specification

We estimate a two-way fixed effects panel specification relating the Red List Index to economic complexity:

$$\log RLI_{it} = \beta_0 + \beta_1 ECI_{it} + \beta_2 \log GDP_{it} + \beta_3 \log Popden_{it} + \beta_4 HDI_{it} + \theta_i + \delta_t + \varepsilon_{it}, \tag{3.1}$$

where RLI_{it} is the Red List Index and ECI_{it} is the standardized Economic Complexity Index. The terms θ_i and δ_t denote country and period fixed effects, absorbing all time-invariant country characteristics (including geography, scale, and baseline biodiversity) and global shocks common to all countries.

The control set is intentionally restricted to variables that separate economic structure from scale and development. GDP per capita (GDP_{it}, constant 2017 PPP US dollars) captures the level of economic activity; population density ($Popden_{it}$) proxies direct anthropogenic pressure on habitat (Cincotta et al. 2000); and the Human Development Index (HDI_{it}) captures broader institutional and development correlates of conservation capacity. Additional covariates (including land-use intensity, protected areas, and production structure) are introduced individually in Section 4.3 to assess robustness and mechanisms. Standard errors are clustered at the country level. The coefficient β_1 is therefore identified from within-country variation in economic complexity conditional on income and other controls.

We complement the two-way fixed effects specification with a long-difference estimator that exploits medium-run variation in both biodiversity outcomes and economic complexity. This approach is motivated by two considerations. First, the Red List Index evolves slowly because species' extinction-risk categories are reassessed only at multi-year intervals, implying limited meaningful high-frequency variation. Second, the theoretical framework emphasizes structural changes in production and their persistent ecological consequences, which are more naturally captured by medium-run shifts in economic complexity than by short-run fluctuations. The long-difference specification therefore reduces attenuation arising from measurement noise and provides a robustness check based on lower-frequency variation. The estimator relates changes in extinction risk over the full sample period to corresponding changes in economic complexity and the control variables:

$$\Delta \log RLI_i = \beta_1 \Delta ECI_i + \beta_2 \Delta \log GDP_i + \beta_3 \Delta \log Popden_i + \beta_4 \Delta HDI_i + \Delta \varepsilon_i, \tag{3.2}$$

where Δ denotes the change between the first and last period in the panel. This transformation removes all time-invariant country characteristics, including geography, baseline biodiversity endowments, and historical development trajectories. The long-difference estimator can thus be interpreted as a medium-run analogue of the fixed-effects specification: whereas the latter exploits within-country deviations around time means, the former compares initial and final outcomes over a twenty-five-year horizon. Consistency of the two approaches strengthens the interpretation that the relationship between economic complexity and extinction risk is not driven by short-run noise or transitory shocks, but reflects persistent changes in economic structure.

To strengthen causal interpretation, we implement an instrumental-variables strategy based on the Complexity Outlook Index (COI), which measures a country's scope for diversification into more complex products given its position in the global product space (Hausmann et al. 2014, Hidalgo 2021). The first stage is:

$$ECI_{it} = \pi_0 + \pi_1 COI_{it} + \pi_2 \log GDP_{it} + \pi_3 \log Popden_{it} + \pi_4 HDI_{it} + \theta_i + \delta_t + u_{it}. \tag{3.3}$$

The second stage replaces ECI_{it} in Equation (3.1) with its fitted value.

The exclusion restriction requires that diversification opportunities affect extinction risk only through realized economic complexity. This restriction is motivated by three features of the COI. First, it is constructed entirely from the structure of international trade and the product space and contains no ecological or land-use information. Second, it is a slow-moving object determined by accumulated capabilities, reflecting historical production patterns rather than contemporaneous environmental conditions. Third, conditional on income, population density, and human development, it captures variation in feasible diversification that is orthogonal to direct drivers of biodiversity loss. As a further check, we estimate specifications using lagged COI, which is predetermined with respect to current extinction outcomes by construction. Formal validity diagnostics are reported in Appendix B.

4 Results

This section presents the paper's three empirical results. We first test the within-country implication of the model linking economic complexity to extinction risk, and assess its causal interpretation using an instrumental-variables strategy. We then investigate the international transmission channel using a bilateral matrix of extinction-risk footprints, and evaluate the range-overlap mechanism by relating imported biodiversity pressure to changes in domestic extinction risk.

4.1 Economic complexity and biodiversity threat

Conditional on income, economic complexity is negatively related to the Red List Index: more complex economies face higher aggregate extinction risk. Table 1 reports the estimates. The columns add controls sequentially within each design: economic complexity alone (columns 1 and 4); economic complexity with log GDP per capita (columns 2 and 5); and the full set of controls including log population density and the Human Development Index (columns 3 and 6). The coefficient on the Economic Complexity Index is stable across specifications. In the two-way fixed effects design, a one-standard-deviation increase in complexity lowers the Red List Index by between 0.8 and 1.2 percent (columns 1–3), with the coefficient essentially unchanged as controls are added. The long-difference design returns a larger coefficient, between 1.3 and 2.2 percent, again stable across specifications (columns 4–6).

Table 1: Economic complexity and the Red List Index

	Country + period FE			Long difference		
	(1)	(2)	(3)	(4)	(5)	(6)
Economic Complexity Index	-0.008**	-0.010***	-0.012***	-0.013**	-0.018***	-0.022***
	(0.004)	(0.004)	(0.004)	(0.006)	(0.006)	(0.006)
log GDP per capita		0.002	-0.020**		0.004	-0.047***
		(0.005)	(0.008)		(0.007)	(0.012)
log Population density			-0.030***			-0.054***
			(0.011)			(0.013)
Human Development Index			0.134**			0.286***
			(0.061)			(0.106)
Country FE	Yes	Yes	Yes	Diff.	Diff.	Diff.
Period FE	Yes	Yes	Yes			
Within R^2	0.404	0.405	0.418	0.023	0.042	0.141
Observations	1,125	1,078	1,000	181	165	129

Notes: This table estimates the within-country fixed-effects model, Equation (3.1) (columns 1–3), and its long difference between the first and last periods, Equation (3.2) (columns 4–6). Columns add controls sequentially: ECI alone (columns 1 and 4); ECI and log GDP per capita (columns 2 and 5); the full set of controls (columns 3 and 6). The dependent variable is the log Red List Index. The Economic Complexity Index is standardized, so its coefficient is the effect of a one-standard-deviation change. Standard errors in parentheses are cluster-robust by country (columns 1–3) and heteroskedasticity-robust (columns 4–6). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The coefficient on the Economic Complexity Index is robust across specifications. Adding income per capita, population density, and the Human Development Index as controls does not displace it; if anything, the estimate strengthens slightly in magnitude. Complexity therefore captures a distinct margin and is not standing in for any of these standard development covariates.

The magnitudes are modest because the Red List Index moves very little within countries over time. Its within-country standard deviation is roughly one-fifth of its overall standard deviation, since species' extinction-risk categories are reassessed only every five to ten years. The complexity association is therefore precisely estimated but small in level terms. Its importance is qualitative: it reverses the sign that the decoupling hypothesis would predict.

The direction of this within-country result is harder to reconcile with the cross-sectional biodiversity–income literature, which finds an inverted-U between income and species threat and suggests that further growth may ease pressure on biodiversity (Naidoo & Adamowicz 2001, Asafu-Adjaye 2003, Dietz &

Adger 2003, Mills & Waite 2009); it also differs in direction from the complexity-and-carbon literature, which finds that complexity lowers territorial emissions (Romero & Gramkow 2021, Mealy & Teytelboym 2022, Boleti et al. 2021). The closest precedent in our direction is Neagu (2020), who finds that complexity raises the aggregate ecological footprint across 48 complex economies. Our panel evidence is complementary: it isolates the effect at the biodiversity margin and uses within-country variation, addressing the cross-country confounding that has long been a concern in this literature (Stern 2004).

4.2 Instrumenting economic complexity

To strengthen the causal interpretation of the baseline estimates, we implement the instrumental-variables strategy described in Section 3.2. The Complexity Outlook Index (COI) is a strong predictor of subsequent economic complexity: the first-stage F-statistic equals 31 (11 when the instrument is lagged one period), comfortably exceeding conventional weak-instrument thresholds. The second-stage estimates remain negative and statistically significant, confirming the baseline relationship.

Two violations of the exclusion restriction are most plausible: the COI could correlate with institutional capacity (rule of law, the enforcement of conservation policy) that itself shapes extinction risk, or with capital accumulation that finances both productive diversification and land-modifying infrastructure such as roads, ports, and mining facilities. Country and period fixed effects absorb time-invariant institutional differences across countries and common global shocks to capital flows, and the contemporaneous controls for log GDP per capita, log population density, and the Human Development Index capture changes in income, scale, and broader development capacity. The COI's predictive content for complexity should therefore be orthogonal to these channels, leaving the exclusion restriction tenable.

Two additional exercises corroborate this argument. First, augmenting the structural equation with the COI alongside the Economic Complexity Index yields a coefficient on the instrument that is small and statistically insignificant, consistent with the absence of a direct effect of diversification opportunities on extinction risk beyond their influence through realized economic complexity. Second, a falsification exercise using a lead of the COI produces no significant association with current biodiversity outcomes, providing no evidence of anticipation effects or pre-existing trends. Together, these results support the identifying assumption that the COI affects extinction risk only through its effect on economic complexity. Appendix B reports the complete set of first-stage and falsification diagnostics.

Table 2 reports the instrumental-variables estimates. The OLS coefficient on the Economic Complexity Index is -0.012. Instrumenting with the contemporaneous Complexity Outlook Index yields an estimate of -0.016, while using the predetermined one-period-lagged instrument produces a larger estimate of -0.038. All estimates are negative, economically comparable in magnitude, and statistically significant at conventional levels.

The relevant comparison is between the OLS and IV estimates. Although instrumental-variables estimators are less efficient than OLS and therefore produce wider confidence intervals, both instrumented specifications remain statistically indistinguishable from the baseline estimate. Correcting for potential endogeneity therefore does not alter either the sign or the economic magnitude of the relationship between economic complexity and extinction risk. If anything, the larger IV coefficients suggest that attenuation or measurement error biases the OLS estimate toward zero rather than away from it. Taken together with the first-stage strength and the exclusion-restriction diagnostics, the IV results reinforce the interpretation that the negative association between economic complexity and the Red List Index

Table 2: Instrumental-variables estimates of the effect of complexity on extinction risk

	(1) OLS	(2) IV: COI	(3) IV: COI (lag)
Economic Complexity Index	-0.012***	-0.016*	-0.038*
	(0.004)	(0.009)	(0.020)
log GDP per capita	-0.020**	-0.020**	-0.016*
	(0.008)	(0.008)	(0.009)
log Population density	-0.030***	-0.031***	-0.027**
	(0.011)	(0.011)	(0.012)
Human Development Index	0.134**	0.143**	0.189**
	(0.061)	(0.058)	(0.079)
First-stage F		31.4	11.3
Observations	999	999	863

Notes: Estimates use the six-period (5-year-window) panel, 1993–2019, with country and period fixed effects and the controls of Equation (3.1) (income per capita, population density, the Human Development Index; coefficients omitted). Column 1 is OLS; columns 2–3 instrument the standardized Economic Complexity Index with the Complexity Outlook Index (Hausmann et al. 2014, Hidalgo 2021), contemporaneous and lagged one period respectively. Standard errors clustered by country in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

reflects a causal effect rather than reverse causality or omitted-variable bias.

4.3 Domestic channels and robustness

The baseline relationship between economic complexity and extinction risk is robust to the checks that matter most in a short within-country panel. This subsection also examines the domestic mechanisms that could in principle generate this relationship and shows that they do not account for the estimated effect.

Sensitivity to controls. A natural concern is that economic complexity proxies for omitted domestic determinants of habitat loss. We address this by augmenting the preferred specification, which already includes income, population density, and the Human Development Index, with additional controls one at a time. Each variable is selected on theoretical grounds rather than for exhaustive conditioning. We consider conservation effort (protected-area coverage), land-use composition (forest, agricultural, and urban shares), production intensity (energy use, fertilizer and pesticide application, and agricultural productivity), and sectoral structure (manufacturing, services, and agricultural value added).

Table 3 reports the corresponding estimates. The coefficient on economic complexity is highly stable, remaining between -0.008 and -0.013 and statistically significant in all specifications. No measure of domestic land use, agricultural intensity, or sectoral composition meaningfully attenuates the effect. Agricultural inputs and productivity variables produce modest attenuation in smaller samples, but the coefficient remains negative and significant throughout. Likewise, conditioning on the sectoral composition of value added leaves the estimate essentially unchanged, indicating that economic complexity is not proxying for the dominance of specific sectors but rather for the sophistication of production.

These results clarify two distinct questions. First, domestic land use is an important determinant of

Table 3: Stability of the complexity coefficient under additional controls

	(1) Pref.	(2) +Prot. area	(3) +FDI	(4) +Forest	(5) +Cropland	(6) +Urban	(7) +Energy	(8) +Fert.	(9) +Pest.	(10) +AgTFP	(11) +Manuf.	(12) +Serv.	(13) +Agric.
Economic Complexity Index	-0.012***	-0.010**	-0.012***	-0.012***	-0.012***	-0.012***	-0.010***	-0.008**	-0.009**	-0.010**	-0.013***	-0.012***	-0.012***
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Protected-area share		0.134*											
		(0.073)											
FDI (% of GDP)			-0.000										
			(0.000)										
Forest share				0.287**									
				(0.140)									
Agricultural land share					0.000								
					(0.000)								
Urban land share						-0.099							
						(0.061)							
log Energy consumption							-0.012**						
							(0.005)						
log Fertilizer (N)								0.004*					
								(0.002)					
log Pesticides									0.002				
									(0.003)				
Agricultural TFP (log)										-0.000			
										(0.007)			
log Manufacturing VA share											0.006		
											(0.004)		
log Services VA share												0.008	
												(0.007)	
log Agriculture VA share													-0.001
													(0.005)
Within R^2	0.418	0.408	0.420	0.444	0.419	0.418	0.448	0.432	0.427	0.415	0.416	0.392	0.418
Observations	1,000	810	997	993	995	993	992	758	718	920	930	946	970

Notes: Column 1 reports the preferred within-country specification, Equation (3.1) (country and period fixed effects, log income per capita, log population density, and the Human Development Index). Columns 2–13 add, one at a time, the additional control named at the head of the column; only the coefficient on the standardized Economic Complexity Index (row 1) and on the added control (its own row) is shown, with the base controls suppressed for compactness. Standard errors clustered by country in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

biodiversity outcomes: forest cover, the most direct measure of habitat availability, enters positively and significantly ($p < 0.05$), implying that countries with greater forest endowment exhibit lower extinction risk. Second, however, the effect of economic complexity does not appear to operate through domestic land-use margins.

Sample, functional form, and frequency. Three additional robustness checks, reported in Appendix D, confirm the stability of the baseline estimate. First, restricting the sample to the balanced panel of countries observed in all six periods leaves the coefficient essentially unchanged at -0.013 ($p < 0.01$). Second, allowing for a quadratic in income does not generate evidence of a biodiversity Environmental Kuznets Curve: neither the linear nor the quadratic income term is statistically significant, and the coefficient on economic complexity remains stable, consistent with the view that the EKC pattern in cross-country data primarily reflects income-level differences rather than within-country dynamics (Stern 2004, Vincent 1997). Third, re-aggregating the data into three longer periods (1993–95, 2005–07, and 2017–19) yields a slightly larger estimate of -0.015 ($p < 0.01$), indicating that the result is not sensitive to the temporal frequency of observation.

Across all specifications, the coefficient on economic complexity remains tightly clustered around its preferred estimate of -0.012 and retains statistical significance, indicating that the baseline relationship is robust to alternative sample restrictions, functional form assumptions, and temporal aggregation.

First-link tests for the domestic channel. Corollary 2.1 implies that the effect of economic complexity on extinction risk is not mediated by domestic habitat conversion. A necessary condition for any such mediation channel is that economic complexity systematically affects the candidate mediators. We therefore test this first link directly by estimating, for each potential channel X_{it},

$$X_{it} = \pi_0 + \pi_1 ECI_{it} + \pi_2 \log GDP_{it} + \pi_3 \log Popden_{it} + \pi_4 HDI_{it} + \theta_i + \delta_t + u_{it}, \tag{4.1}$$

on the six-period panel with country-clustered standard errors. The coefficient π_1 captures whether economic complexity is associated with systematic changes in the proposed domestic channels, conditional on income, population density, and human development. A coefficient statistically indistinguishable from zero rules out that channel as a first-stage mechanism linking complexity to extinction risk.

We consider variables that span the main theoretically plausible domestic transmission margins. These include land-use outcomes (forest cover, agricultural land, urban land, and the rate of deforestation), proximate drivers of habitat pressure (agricultural productivity, energy consumption, foreign direct investment, and protected-area coverage), and sectoral composition (value-added shares of manufacturing, services, and agriculture). Together, these variables exhaust the principal domestic pathways through which changes in economic structure could be expected to affect biodiversity outcomes.

Ten of the eleven coefficients reported in Table 4 are statistically indistinguishable from zero at conventional levels ($p > 0.10$). Conditional on income and the baseline controls, economic complexity does not systematically affect forest cover, agricultural land, urban land, the rate of deforestation, agricultural productivity, energy consumption, foreign direct investment, protected-area coverage, or the sectoral shares of services and agriculture. The absence of significant effects on these variables implies that the first-stage condition for domestic mediation is not satisfied: these channels do not respond to variation in economic complexity and therefore cannot explain the within-country complexity–RLI relationship.

Table 4: First-link tests: effect of economic complexity on candidate domestic channels

Dependent variable	ECI coefficient	Std. error	Obs.
Forest share	0.0001	(0.0022)	997
Agricultural land share	0.0058	(0.4936)	999
Urban land share	0.0045	(0.0034)	997
Deforestation rate	0.0950	(0.6519)	993
Agricultural TFP (log)	0.0347	(0.0292)	920
log Energy consumption	0.0713	(0.0513)	997
FDI (% of GDP)	1.8905	(1.2076)	998
Protected-area share	0.0025	(0.0025)	808
log Manufacturing VA share	0.0802^{*}	(0.0444)	934
log Services VA share	0.0024	(0.0193)	950
log Agriculture VA share	0.0160	(0.0350)	974

Notes: Each row reports the coefficient on the standardized Economic Complexity Index from Equation (4.1), with the named variable as the dependent variable. All regressions include country and period fixed effects, log GDP per capita, log population density, and the Human Development Index. Standard errors clustered by country in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The only statistically detectable effect of economic complexity is on the manufacturing value-added share, which increases modestly with complexity (0.08, $p = 0.07$). This is consistent with the compositional mechanism embedded in the model (Lemma 2.1): as productive capabilities expand, labor is reallocated toward manufacturing and away from land-intensive activities. However, the relevant empirical question is whether this margin mediates the biodiversity relationship. Table 3 shows that it does not. Including the manufacturing share in the baseline RLI specification leaves the coefficient on economic complexity essentially unchanged at -0.013 (significant at the 1% level). Thus, although manufacturing responds to complexity, its magnitude is insufficient to account for the observed relationship with extinction risk.

Overall, the evidence indicates a systematic failure of first-link effects across the set of plausible domestic channels. We therefore reject the domestic mediation channel in the sense of sequential ignorability. The within-country association between economic complexity and biodiversity loss is not transmitted through standard domestic land-use, production-intensity, or sectoral-composition margins.

Why, then, do domestic channels fail to explain the effect? A natural reading is that the biodiversity cost of a complex economy is not borne mainly at home. As Section 4.4 below shows, complex economies increasingly source their consumption footprint abroad, so the biodiversity pressure associated with complexity is largely invisible in domestic land-use accounts: the mechanism is trade-embodied rather than territorial. This result is less easily reconciled with mechanism-based readings of the biodiversity–income relationship that locate any EKC in domestic land-use efficiency or conservation institutions, and is more consistent with the pollution-haven and Copeland & Taylor (1994, 2004) view, supported empirically by Cole (2004), that complex economies often relocate ecologically intensive production to weaker-regulation partners rather than reform it at home.

4.4 The geography of the extinction-risk footprint

The absence of a domestic mediation channel suggests that the biodiversity consequences of economic complexity are not primarily borne within national borders. Agricultural expansion, forest loss, input intensity, and agricultural productivity do not respond systematically to economic complexity, and conditioning on these variables does not alter the baseline relationship between complexity and extinction risk (Section 4.3). This pattern indicates that domestic land-use adjustments are not the main transmission margin.

The implication is that the biodiversity cost of complexity is geographically displaced. Rather than being eliminated, extinction pressure is embodied in production networks and reallocated across countries through trade. In supplier economies, this displacement is associated with land conversion for export-oriented agriculture (including plantations and aquaculture), extraction activities integrated into global value chains, and the associated chemical and infrastructural pressures (fertilizer and pesticide runoff, mining impacts, and transport infrastructure). The extinction-risk footprint framework (Irwin et al. 2022, Lenzen et al. 2012, Moran & Kanemoto 2017) aggregates these channels into a bilateral accounting system that assigns biodiversity pressure embodied in consumption to its ultimate geographic origin.

We examine this mechanism using the 2013 cross-sectional matrix of extinction-risk footprints:[1]

$$y_i = \alpha + \delta_1 ECI_i + \delta_2 \log GNI_i + u_i, \tag{4.2}$$

where y_i is, in turn, the log imported footprint, the log domestic footprint, the foreign share of the total footprint, and the net import position (Table 5); $\log GNI_i$ is the log of Gross National Income per capita. We use GNI rather than GDP here because the outcomes are consumption-based: GNI includes net factor income from abroad and therefore better captures consumption capacity than territorial GDP, which is the convention in the multi-regional input–output literature (Lenzen et al. 2012, Moran & Kanemoto 2017, Irwin et al. 2022). This specification is the empirical counterpart of Proposition II, which predicts that both the foreign share of the consumption footprint and the net imported extinction risk rise with complexity. The test isolates whether economically complex countries systematically differ in the geography of the biodiversity pressure associated with their consumption.

Table 5: Complexity, income, and the geography of the extinction-risk footprint

	(1) log Imported	(2) log Domestic	(3) Foreign share	(4) Net import
Economic Complexity Index	0.294*	−0.255	0.040***	662.6***
	(0.160)	(0.246)	(0.008)	(230.3)
log GNI per capita	0.002	−0.730***	0.042***	211.1
	(0.138)	(0.235)	(0.008)	(162.7)
Observations	173	173	173	173
R^2	0.028	0.108	0.475	0.158

Notes: Each column estimates the 2013 cross-sectional regression Equation (4.2) for a different outcome: the log imported footprint (1), the log domestic footprint (2), the share of the consumption footprint falling abroad (3), and the net import position (4). "Net import" is imported minus exported extinction-risk footprint, positive for net importers. The Economic Complexity Index is standardized. Heteroskedasticity-robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Two results are central. First, the geographic composition of the biodiversity footprint shifts sharply

[1]The bilateral extinction-risk footprint matrix of Irwin et al. (2022) is currently available only for 2013; a panel version has not yet been constructed.

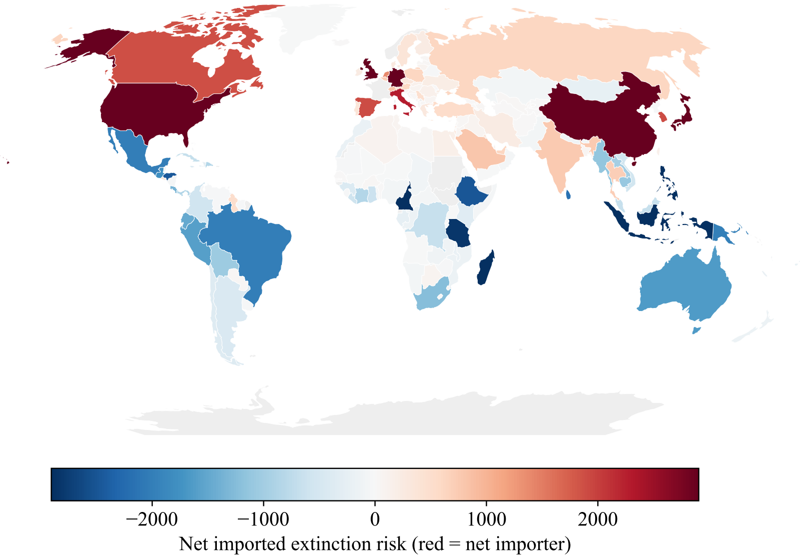


Figure 2: The geography of outsourced extinction risk, 2013.

Notes: Each country is shaded by its net imported extinction risk (the extinction risk its consumption causes abroad minus the risk foreign consumption causes within it). Red denotes net importers (consumption causes more loss abroad than is borne at home) and blue denotes net exporters. Values are computed from the 2013 bilateral extinction-risk footprint matrix (Irwin et al. 2022, Lenzen et al. 2012).

with economic complexity. The foreign share of a country's consumption footprint increases strongly in complexity: a one-standard-deviation rise in the ECI raises this share by approximately four percentage points, and together with income it explains close to half of its cross-country variation (column 3). Second, the net position reverses with complexity: while low-complexity economies are net exporters of extinction risk, high-complexity economies are net importers (column 4).

This pattern is reflected in both margins of the decomposition. More complex economies are associated with a larger imported footprint (column 1), while domestic footprint declines with income (column 2). The joint implication is a reallocation of biodiversity pressure across space: high-income, high-complexity economies generate relatively limited domestic extinction risk but increasingly rely on external ecosystems to sustain consumption.

The gradient is economically large across the distribution. Countries in the top tercile of complexity place roughly ninety percent of their consumption footprint abroad and are net importers of extinction risk, whereas those in the bottom tercile externalize about seventy-four percent and are net exporters. Figure 2 summarizes the resulting geography: North America, Western Europe, and East Asia are systematic net importers, while Latin America, sub-Saharan Africa, and Southeast Asia are net exporters. This spatial pattern complements the within-country results: although biodiversity risk rises with complexity domestically, an increasing share of the associated pressure is displaced onto trading partners' ecosystems.

This evidence aligns with the multi-regional input–output literature documenting the displacement of biodiversity and ecological pressure from high-income to biodiversity-rich economies (Lenzen et al. 2012, Moran & Kanemoto 2017, Marques et al. 2019, Wilting et al. 2017, Chaudhary & Kastner 2016, Irwin et al. 2022). The contribution here is to show that this displacement is not explained by income alone: even conditional on income, economic complexity is a key structural determinant of the foreign share of the extinction-risk footprint.

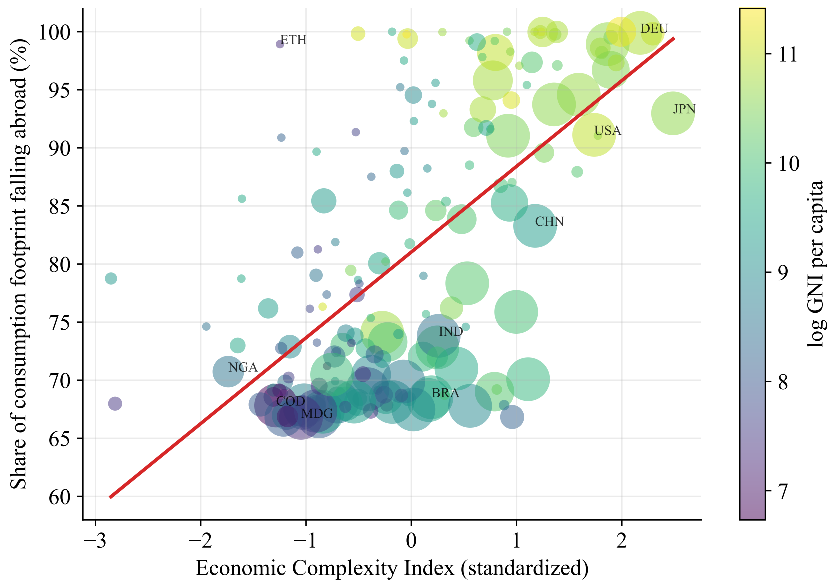


Figure 3: Economic complexity and the share of the consumption footprint that falls abroad

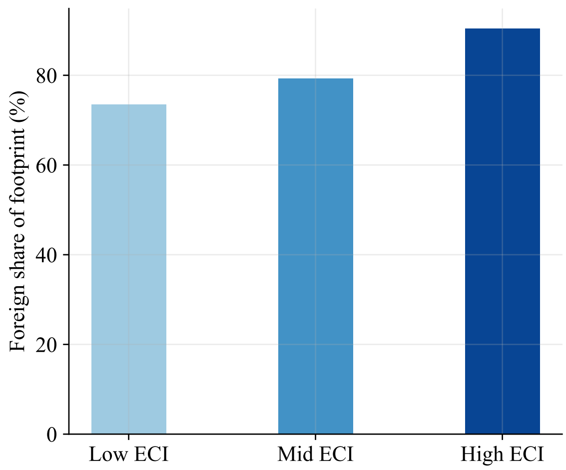
Notes: Each point is a country in the 2013 cross-section. The vertical axis is the share of a country's consumption footprint that falls on other countries' species; the horizontal axis is the standardized Economic Complexity Index. Point size is proportional to the total consumption footprint and color encodes log GNI per capita. The fitted line corresponds to the foreign-share regression, Equation (4.2).

The leakage gradient is continuous across the entire distribution of economic complexity rather than being driven by a discrete separation between low- and high-complexity economies. Figure 3 plots, for each country in the 2013 cross-section, the share of its consumption footprint embodied abroad against its standardized Economic Complexity Index. The relationship is steep and tightly estimated: a one-standard-deviation increase in complexity raises the foreign share by approximately four percentage points (Table 5, column 3), and together with income it accounts for nearly half of the cross-country variation in this outcome.

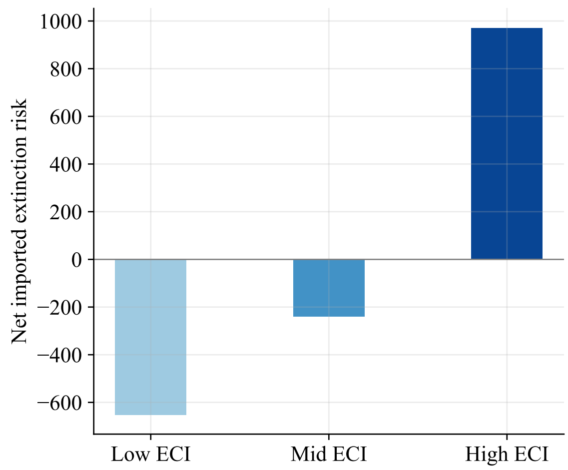
The pattern is primarily organized along the complexity dimension rather than income. In the bivariate representation, income (indicated by color shading) exhibits comparatively weak structure relative to the strong monotonic gradient in complexity: high-income but resource-intensive economies lie systematically below the fitted relationship, while lower-income but more complex economies lie above it. Complexity therefore captures a dimension of production structure that is distinct from income and more closely aligned with the geography of embodied biodiversity pressure.

Aggregating the data into terciles yields a consistent and more transparent summary. Figure 4, panel (a), reports mean foreign shares by complexity group. Even the bottom tercile externalizes roughly three-quarters of its consumption footprint, but this share rises monotonically across the distribution, reaching close to ninety percent in the top tercile. Panel (b) reports the corresponding net position in extinction-risk footprints. Low-complexity economies are net exporters of biodiversity pressure, whereas high-complexity economies are net importers.

Taken together, the two panels restate the regression evidence in Table 5 in distributional form. As economic complexity increases, a growing share of consumption-related biodiversity loss is displaced abroad, and the role of countries shifts from net suppliers to net absorbers of global extinction risk.



(a) Share of footprint falling abroad



(b) Net imported extinction risk

Figure 4: The extinction-risk footprint by tercile of economic complexity

Notes: Countries are grouped into terciles of the standardized Economic Complexity Index. Panel (a) shows the mean share of the consumption footprint that falls abroad; panel (b) shows the mean net imported extinction risk (positive values denote net importers). Both rise monotonically with complexity.

4.5 Imports from biome-sharing regions and own-country decline

The previous two results establish complementary facts. Within countries, greater economic complexity is associated with a decline in the Red List Index. Across countries, more complex economies externalize an increasing share of their biodiversity footprint through trade. The theoretical framework links these findings through the species-range overlap matrix Ω. From Equation (2.8), the effect of Home's complexity on its biodiversity index operates not only through domestic habitat conversion but also through the foreign component $-\omega_{HF}\, m_F\, (dh_F/d\kappa_H)$, which captures habitat conversion induced abroad by Home's consumption and transmitted back through shared species ranges. The empirical implication is that countries importing a larger biodiversity footprint from biome-sharing regions should experience a larger subsequent decline in their own Red List Index.

We test this prediction by estimating

$$\Delta \log RLI_i = \alpha + \beta_1 B_i + \gamma \Delta ECI_i + \pi' \Delta \mathbf{z}_i + \varepsilon_i, \tag{4.3}$$

where Δ denotes the long difference between the 1993–95 and 2018–19 periods, B_i denotes the bridge variable, and $\Delta \mathbf{z}_i$ includes the long-difference controls for log GDP per capita, log population density, and the Human Development Index. The theoretical prediction is $\beta_1 < 0$.

Table 6 considers three measures of the bridge variable. Column 1 uses the total imported extinction-risk footprint, $F_i^{\text{imp}} = \sum_{j \neq i} F_{j \to i}$, where $F_{j \to i}$ denotes the extinction-risk footprint generated in source country j by consumption in country i. Column 2 replaces this measure with the overlap-weighted imported footprint, $F_i^{\omega} = \sum_{j \neq i} \omega_{ij} F_{j \to i}$, which is the empirical counterpart of the range-overlap mechanism in the model. Column 3 uses the overlap share, $\sigma_i = F_i^{\omega}/F_i^{\text{imp}}$, while controlling for the total imported footprint, thereby isolating the composition of imports from their overall scale.

Table 6: Imports from biome-sharing regions and own-country extinction-risk decline

	(1) Direct	(2) ω-weighted	(3) Overlap share
log Imported footprint (F^{imp})	-0.011***		-0.011***
	(0.002)		(0.002)
log ω-weighted imported (F^{ω})		-0.010***	
		(0.002)	
Overlap share of imports (σ)			-0.153**
			(0.070)
Δ ECI	-0.016***	-0.015***	-0.015***
	(0.005)	(0.005)	(0.005)
Δ log GDP per capita	-0.040***	-0.042***	-0.041***
	(0.011)	(0.011)	(0.011)
Δ log Population density	-0.050***	-0.046***	-0.046***
	(0.012)	(0.012)	(0.012)
Δ HDI	0.212**	0.229**	0.203**
	(0.100)	(0.103)	(0.100)
R^2	0.336	0.331	0.359
Observations	118	118	118

Notes: The dependent variable is the long-difference $\Delta \log RLI$ for each country between the 1993–95 and 2018–19 windows. The bridge variables are built from the 2013 bilateral extinction-risk footprint matrix (Irwin et al. 2022). "ω-weighted imported" multiplies imported footprint by the within-continent overlap proxy for ω_{ij}; "Overlap share" is the share of imported footprint sourced from countries on the same continent. Heteroskedasticity-robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The evidence strongly supports the bridge mechanism. A larger imported extinction-risk footprint is associated with a greater decline in the Red List Index (-0.011, $p < 0.01$; column 1). Weighting imports by species-range overlap produces a coefficient of similar magnitude (-0.010, $p < 0.01$; column 2), indicating that biodiversity pressure originating in regions sharing species with the importing country is particularly predictive of domestic biodiversity decline. The overlap share likewise enters negatively and significantly (-0.153, $p < 0.05$; column 3), implying that countries sourcing a larger fraction of their biodiversity footprint from biome-sharing regions experience steeper declines in their own Red List Index, even after conditioning on the overall scale of imported biodiversity pressure.

Across all specifications, the coefficient on economic complexity remains stable, between -0.015 and -0.016 ($p < 0.01$). The bridge variables therefore contribute explanatory power beyond economic complexity itself, providing direct empirical support for the species-range overlap mechanism developed in the theoretical framework. Appendix C describes the construction of the overlap matrix Ω and the bridge variables, and reports robustness to alternative measures of species-range overlap.

Spatial dependence. Corollary 2.2 predicts that biodiversity indices are positively spatially dependent because species' ranges extend across national borders. The data strongly support this prediction. Extinction risk clusters geographically: the contiguity Moran's I of the Red List Index is 0.54, and the spatial autoregressive coefficient is positive and highly significant, consistent with the fact that neighboring countries often share ecosystems and that a species classified as threatened carries the same conservation status throughout its range (Pandit & Laband 2007).

The question is whether this spatial dependence accounts for the estimated relationship between economic complexity and extinction risk. Table 7 shows that it does not. First, absorbing common regional shocks with continent-by-period fixed effects leaves the coefficient on economic complexity negative and statistically significant, although somewhat attenuated (-0.007, $p = 0.05$). Second, we estimate the spatial autoregressive specification

$$\Delta \log RLI_i = \rho \sum_j W_{ij} \Delta \log RLI_j + \beta_1 \Delta ECI_i + \mathbf{z}_i' \gamma + \varepsilon_i, \tag{4.4}$$

where W_{ij} denotes the row-standardized queen-contiguity proxy for the theoretical spatial weight matrix introduced in Corollary 2.2, and ρ measures spatial dependence. The model is estimated by spatial two-stage least squares, instrumenting the spatial lag with the spatial lags of the exogenous regressors.

Table 7: Spatial robustness of the complexity–extinction-risk association

	(1) Continent×period FE	(2) Spatial lag, changes
Economic Complexity Index	-0.007^{**}	-0.011
	(0.003)	(0.007)
Spatial lag ρ	—	0.609^{***}
		(0.150)
Country FE	✓	—
Period FE	✓	—
Estimator	Within (FE)	Spatial 2SLS
Specification	Levels (panel)	Long difference
Observations	878	119

Notes: The dependent variable is the log Red List Index (column 1) and its long difference (column 2). Column 1 adds continent-by-period fixed effects to the preferred within-country specification, Equation (3.1), with cluster-robust (by country) standard errors. Column 2 estimates the spatial autoregressive model Equation (4.4) by spatial two-stage least squares with a row-standardized queen-contiguity weight matrix, with heteroskedasticity-robust standard errors; ρ is the spatial-lag parameter. The contiguity Moran's I of the Red List Index is 0.54. Both specifications include income per capita, population density, and the Human Development Index (coefficients omitted). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Accounting explicitly for cross-country dependence leaves the substantive conclusion unchanged. The coefficient on economic complexity remains negative and close to its baseline magnitude (-0.011 versus -0.012), although the smaller long-difference sample reduces precision ($p \approx 0.11$). At the same time, the spatial autoregressive parameter is large and positive ($\rho = 0.61$), confirming that biodiversity outcomes are strongly interconnected across neighboring countries. The evidence therefore supports the final prediction of the theoretical framework: biodiversity loss is spatially transmitted across countries, yet this spatial dependence does not explain away the within-country relationship between economic complexity and extinction risk. Taken together, the results support a coherent interpretation. Economic complexity increases extinction risk within countries, not because it intensifies domestic habitat conversion, but because it increasingly displaces biodiversity pressure through international trade, where it feeds back through shared species ranges.

5 Discussion

The evidence points to a common conclusion: economic complexity changes the geography of biodiversity loss rather than eliminating it. More complex economies experience rising extinction risk within countries even though this relationship is not explained by domestic land use. Instead, complexity increasingly shifts the biodiversity footprint of consumption toward foreign ecosystems, where the resulting habitat conversion feeds back through species whose ranges cross national borders. The central implication is that sophistication alters where biodiversity pressure is realized, not whether it is generated.

This perspective helps reconcile several strands of the literature. Cross-sectional studies of biodiversity and development often report an Environmental Kuznets Curve, suggesting that biodiversity pressure eventually declines with income (Naidoo & Adamowicz 2001, Asafu-Adjaye 2003, Dietz & Adger 2003, Mills & Waite 2009). An early and influential skeptical voice within this literature is Dietz & Adger (2003), who find no robust inverted-U for biodiversity and argue that intentional conservation effort, rather than rising income alone, is the binding margin for protecting species. Mills & Waite (2009) extend this skepticism using global IUCN Red List data, showing that the inverted-U is weak and specification-sensitive across taxa, and that the apparent gains from rising income may be offset by globalization-related pressures on biodiversity in trading partners. Our within-country evidence is less supportive of an EKC interpretation and instead echoes Dietz & Adger (2003), Mills & Waite (2009), and the caution raised by Stern (2004) for environmental Kuznets curves more generally: cross-country comparisons can confound genuine environmental improvement with the spatial relocation of environmental pressure. The displacement mechanism that Mills & Waite (2009) anticipate verbally is exactly what we identify and measure. At the same time, our results complement the economic complexity literature. Whereas complexity has repeatedly been associated with lower territorial carbon emissions and cleaner production (Romero & Gramkow 2021, Mealy & Teytelboym 2022, Stojkoski et al. 2023, Boleti et al. 2021), biodiversity follows a different logic because habitat conversion is spatially localized and embodied in traded goods. A country may successfully decarbonize production at home while continuing to drive habitat conversion in biodiversity-rich trading partners. Our findings therefore do not contradict the complexity literature; rather, they show that biodiversity responds differently from pollutants whose damages are global and whose accounting remains largely production based.

The mechanism is equally informative. The absence of evidence for domestic mediation indicates that the biodiversity consequences of economic complexity are not primarily transmitted through changes in domestic land use, agricultural intensification, conservation effort, or sectoral composition. Instead, the evidence is more consistent with the pollution-haven logic of Copeland & Taylor (1994, 2004), recently revisited for trade-driven leakage by Holladay et al. (2018), Abman et al. (2026), and Carattini et al. (2026): as economies become more sophisticated, environmentally intensive production is increasingly relocated across borders. In the context of biodiversity, however, relocation does not sever the ecological connection between importing and exporting countries. Because species ranges frequently span national boundaries, habitat conversion abroad remains relevant for the conservation status of species occurring at home. The bridge analysis provides direct evidence for this mechanism by showing that imports sourced from biome-sharing regions are associated with larger subsequent declines in domestic biodiversity.

These findings also extend the multi-regional input–output literature on biodiversity footprints (Lenzen et al. 2012, Moran & Kanemoto 2017, Marques et al. 2019, Wilting et al. 2017, Chaudhary & Kastner 2016, Irwin et al. 2022) and complement microeconomic evidence on biodiversity–trade trade-offs at the producer level (Dalheimer et al. 2024). Previous work has established that wealthy economies externalize a substantial share of the biodiversity pressure embodied in their consumption. Our results identify economic complexity as an independent structural determinant of that displacement, conditional on income, and link the geography of embodied biodiversity loss to observed changes in countries' Red List Indices through shared species ranges.

Two recent contributions warrant direct comparison. Wiebe & Wilcove (2025) document outsourced deforestation in a global panel and show that consumption in wealthy economies drives forest loss in their trading partners; our results identify economic complexity, beyond income, as the structural determinant of this displacement and connect it to within-country biodiversity outcomes through species-range overlap. Liang et al. (2021) use US sub-national variation to relate local economic activity to species imperilment and find that economic intensity within a region predicts local biodiversity decline; our cross-country evidence is complementary in scope, identifying the same direction of pressure but isolating the role of national production structure rather than local economic intensity, and showing that the within-country relationship survives accounting for income, population density, and broader development covariates.

The policy implications follow directly from this geography. Production-based and territorial indicators systematically understate the biodiversity consequences of the most complex economies because an increasing share of the associated habitat conversion occurs abroad. As the theoretical framework demonstrates, policies that price only domestic biodiversity damage internalize the diagonal elements of the international externality while leaving the cross-border component unpriced. Efficient conservation therefore requires consumption-based accounting, or equivalent international transfers, that recognize the biodiversity impacts embodied in trade. This conclusion is closely aligned with the wealth- and footprint-based approach advocated by Dasgupta (2021) and suggests that biodiversity policy should increasingly move beyond territorial indicators toward measures that attribute responsibility according to final consumption.

6 Conclusion

This paper asked whether the structure of an economy, beyond its level of income, shapes biodiversity loss. The evidence suggests that it does. Conditional on income, greater economic complexity is associated with higher extinction risk within countries. This relationship is not explained by domestic land use or production intensity. Instead, economically complex countries increasingly externalize the biodiversity footprint of their consumption through international trade, becoming net importers of extinction risk. Economic complexity therefore changes the geography of biodiversity loss rather than reducing it.

The paper makes three contributions. First, it introduces economic complexity as a structural determinant of biodiversity loss, complementing a literature that has focused primarily on income. Second, it develops a trade model with cross-border species ranges that reconciles the three empirical findings: the within-country decline in biodiversity, the absence of domestic mediation, and the international displacement of extinction risk. Third, it shows that countries importing a larger biodiversity footprint from biome-sharing regions experience larger subsequent declines in their own Red List Index, providing direct evidence for the range-overlap mechanism emphasized by the model.

These findings also carry a broader implication for biodiversity policy. Conservation frameworks based solely on territorial indicators systematically understate the biodiversity impacts of economies whose consumption increasingly relies on habitat conversion abroad. Because biodiversity loss is embodied in trade but transmitted through species whose ranges cross national borders, effective conservation requires accounting systems that assign responsibility according to final consumption as well as production. In this sense, the geography of biodiversity loss has outgrown the geography of conventional environmental accounting.

A natural next step is to extend bilateral extinction-risk footprint accounts into a panel dataset. Doing so would allow the relocation of biodiversity pressure to be tracked over time and would permit the range-overlap mechanism identified here to be tested within countries rather than across them. Such data would make it possible to study how structural transformation is reshaping the global geography of biodiversity loss as production networks continue to evolve.

References

Abman, R., Jenkins, H. & Lundberg, C. (2026), 'The limits of cross-border environmental policies: Trade diversion as leakage', *Journal of Environmental Economics and Management* **137**, 103298. doi: https://doi.org/10.1016/j.jeem.2026.103298.

Asafu-Adjaye, J. (2003), 'Biodiversity loss and economic growth: A cross-country analysis', *Contemporary Economic Policy* **21**(2), 173–185. doi: https://doi.org/10.1093/cep/byg003.

BirdLife International & Handbook of the Birds of the World (2019), 'Bird species distribution maps of the world. version 2019.1', http://datazone.birdlife.org/species/requestdis.

Boleti, E., Garas, A., Kyriakou, A. & Lapatinas, A. (2021), 'Economic complexity and environmental performance: Evidence from a world sample', *Environmental Modeling & Assessment* **26**, 251–270. doi: https://doi.org/10.1007/s10666-021-09750-0.

Brooks, T. M., Mittermeier, R. A., Mittermeier, C. G., da Fonseca, G. A. B., Rylands, A. B., Konstant, W. R., Flick, P., Pilgrim, J., Oldfield, S., Magin, G. & Hilton-Taylor, C. (2002), 'Habitat loss and extinction in the hotspots of biodiversity', *Conservation Biology* **16**(4), 909–923. doi: https://doi.org/10.1046/j.1523-1739.2002.00530.x.

Butchart, S. H. M., Akçakaya, H. R., Chanson, J., Baillie, J. E. M., Collen, B., Quader, S., Turner, W. R., Amin, R., Stuart, S. N. & Hilton-Taylor, C. (2007), 'Improvements to the red list index', *PLoS ONE* **2**(1), e140. doi: https://doi.org/10.1371/journal.pone.0000140.

Butchart, S. H. M., Stattersfield, A. J., Bennun, L. A., Shutes, S. M., Akçakaya, H. R., Baillie, J. E. M., Stuart, S. N., Hilton-Taylor, C. & Mace, G. M. (2004), 'Measuring global trends in the status of biodiversity: Red list indices for birds', *PLoS Biology* **2**(12), e383. doi: https://doi.org/10.1371/journal.pbio.0020383.

Carattini, S., Huang, H., Pisch, F. & Singh, T. P. (2026), 'Trade and the scopes of pollution: Evidence from China's world market integration', *Journal of Environmental Economics and Management* **139**, 103378. doi: https://doi.org/10.1016/j.jeem.2026.103378.

Chaudhary, A. & Kastner, T. (2016), 'Land use biodiversity impacts embodied in international food trade', *Global Environmental Change* **38**, 195–204. doi: https://doi.org/10.1016/j.gloenvcha.2016.03.013.

Cincotta, R. P., Wisnewski, J. & Engelman, R. (2000), 'Human population in the biodiversity hotspots', *Nature* **404**(6781), 990–992. doi: https://doi.org/10.1038/35010105.

Cole, M. A. (2004), 'Trade, the pollution haven hypothesis and the environmental kuznets curve: examining the linkages', *Ecological Economics* **48**(1), 71–81. doi: https://doi.org/10.1016/j.ecolecon.2003.09.007.

Copeland, B. R. & Taylor, M. S. (1994), 'North-south trade and the environment', *The Quarterly Journal of Economics* **109**(3), 755–787. doi: https://doi.org/10.2307/2118421.

Copeland, B. R. & Taylor, M. S. (2004), 'Trade, growth, and the environment', *Journal of Economic Literature* **42**(1), 7–71. doi: https://doi.org/10.1257/.42.1.7.

Dalheimer, B., Parikoglou, I., Brambach, F., Yanita, M., Kreft, H. & Brümmer, B. (2024), 'On the palm oil–biodiversity trade-off: Environmental performance of smallholder producers', *Journal of Environmental Economics and Management* **125**, 102975. doi: https://doi.org/10.1016/j.jeem.2024.102975.

Dasgupta, P. (2021), *The Economics of Biodiversity: The Dasgupta Review*, HM Treasury, London.

Díaz, S., Settele, J., Brondízio, E. S., Ngo, H. T. et al. (2019), 'Pervasive human-driven decline of life on earth points to the need for transformative change', *Science* **366**, eaax3100. doi: https://doi.org/10.1126/science.aax3100.

Dietz, S. & Adger, W. N. (2003), 'Economic growth, biodiversity loss and conservation effort', *Journal of Environmental Management* **68**(1), 23–35. doi: https://doi.org/10.1016/s0301-4797(02)00231-1.

Growth Lab at Harvard University (2024), 'The atlas of economic complexity'. Data: https://atlas.hks.harvard.edu.

Hartmann, D., Guevara, M. R., Jara-Figueroa, C., Aristarán, M. & Hidalgo, C. A. (2017), 'Linking economic complexity, institutions, and income inequality', *World Development* **93**, 75–93. doi: https://doi.org/10.1016/j.worlddev.2016.12.020.

Hausmann, R., Hidalgo, C. A., Bustos, S., Coscia, M., Simoes, A. & Yildirim, M. A. (2014), *The Atlas of Economic Complexity: Mapping Paths to Prosperity*, The MIT Press, Cambridge, MA. doi: https://doi.org/10.7551/mitpress/9647.001.0001.

Hausmann, R., Hwang, J. & Rodrik, D. (2007), 'What you export matters', *Journal of Economic Growth* **12**, 1–25. doi: https://doi.org/10.1007/s10887-006-9009-4.

Hidalgo, C. A. (2021), 'Economic complexity theory and applications', *Nature Reviews Physics* **3**, 92–113. doi: https://doi.org/10.1038/s42254-020-00275-1.

Hidalgo, C. A. & Hausmann, R. (2009), 'The building blocks of economic complexity', *Proceedings of the National Academy of Sciences* **106**, 10570–10575. doi: https://doi.org/10.1073/pnas.0900943106.

Hidalgo, C. A., Klinger, B., Barabási, A.-L. & Hausmann, R. (2007), 'The product space conditions the development of nations', *Science* **317**, 482–487. doi: https://doi.org/10.1126/science.1144581.

Hoffmann, M., Hilton-Taylor, C., Angulo, A., Böhm, M., Brooks, T. M., Butchart, S. H. M., Carpenter, K. E., Chanson, J., Collen, B., Cox, N. A. et al. (2010), 'The impact of conservation on the status of the world's vertebrates', *Science* **330**(6010), 1503–1509. doi: https://doi.org/10.1126/science.1194442.

Holladay, J. S., Mohsin, M. & Pradhan, S. (2018), 'Emissions leakage, environmental policy and trade frictions', *Journal of Environmental Economics and Management* **88**, 95–113. doi: https://doi.org/10.1016/j.jeem.2017.10.004.

IPBES (2019), *Global Assessment Report on Biodiversity and Ecosystem Services*, IPBES Secretariat, Bonn, Germany. May 5, 2021.

Irwin, A., Geschke, A., Brooks, T. M., Siikamäki, J., Mair, L. & Strassburg, B. B. N. (2022), 'Quantifying and categorising national extinction-risk footprints', *Scientific Reports* **12**(1), 5861. doi: https://doi.org/10.1038/s41598-022-09827-0.

IUCN (2012), *IUCN Red List Categories and Criteria: Version 3.1*, second edn, IUCN, Gland, Switzerland and Cambridge, UK.

Lenzen, M., Moran, D., Kanemoto, K., Foran, B., Lobefaro, L. & Geschke, A. (2012), 'International trade drives biodiversity threats in developing nations', *Nature* **486**(7401), 109–112. doi: https://doi.org/10.1038/nature11145.

Li, X. & Chen, S. (2024), 'Does trade openness aggravate embodied species loss? evidence from the belt and road countries', *Environmental Impact Assessment Review* **104**, 107343. doi: https://doi.org/10.1016/j.eiar.2023.107343.

Liang, Y., Rudik, I. & Zou, E. (2021), Economic activity and biodiversity in the united states, NBER Working Paper 29357, National Bureau of Economic Research. doi: https://doi.org/10.3386/w29357.

Marques, A., Martins, I. S., Kastner, T., Plutzar, C., Theurl, M. C., Eisenmenger, N., Huijbregts, M. A. J., Wood, R., Stadler, K., Bruckner, M., Canelas, J., Hilbers, J. P., Tukker, A., Erb, K. & Pereira, H. M. (2019), 'Increasing impacts of land use on biodiversity and carbon sequestration driven by population and economic growth', *Nature Ecology & Evolution* **3**(4), 628–637. doi: https://doi.org/10.1038/s41559-019-0824-3.

Maxwell, S. L., Fuller, R. A., Brooks, T. M. & Watson, J. E. M. (2016), 'Biodiversity: The ravages of guns, nets and bulldozers', *Nature* **536**, 143–145. doi: https://doi.org/10.1038/536143a.

Mealy, P. & Teytelboym, A. (2022), 'Economic complexity and the green economy', *Research Policy* **51**, 103948. doi: https://doi.org/10.1016/j.respol.2020.103948.

Mills, J. H. & Waite, T. A. (2009), 'Economic prosperity, biodiversity conservation, and the environmental kuznets curve', *Ecological Economics* **68**, 2087–2095. doi: https://doi.org/10.1016/j.ecolecon.2009.01.017.

Moran, D. & Kanemoto, K. (2017), 'Identifying species threat hotspots from global supply chains', *Nature Ecology & Evolution* **1**, 0023. doi: https://doi.org/10.1038/s41559-016-0023.

Naidoo, R. & Adamowicz, W. L. (2001), 'Effects of economic prosperity on numbers of threatened species', *Conservation Biology* **15**(4), 1021–1029. doi: https://doi.org/10.1046/j.1523-1739.2001.0150041021.x.

Neagu, O. (2020), 'Economic complexity and ecological footprint: Evidence from the most complex economies in the world', *Sustainability* **12**, 9031. doi: https://doi.org/10.3390/su12219031.

Olson, D. M., Dinerstein, E., Wikramanayake, E. D., Burgess, N. D., Powell, G. V. N., Underwood, E. C., D'Amico, J. A., Itoua, I., Strand, H. E., Morrison, J. C., Loucks, C. J., Allnutt, T. F., Ricketts, T. H., Kura, Y., Lamoreux, J. F., Wettengel, W. W., Hedao, P. & Kassem, K. R. (2001), 'Terrestrial ecoregions of the world: A new map of life on earth', *BioScience* **51**(11), 933–938. doi: https://doi.org/10.1641/0006-3568(2001)051[0933:TEOTWA]2.0.CO;2.

Pandit, R. & Laband, D. N. (2007), 'Spatial autocorrelation in country-level models of species imperilment', *Ecological Economics* **60**(3), 526–532. doi: https://doi.org/10.1016/j.ecolecon.2006.07.018.

Pimm, S. L., Jenkins, C. N., Abell, R., Brooks, T. M., Gittleman, J. L., Joppa, L. N., Raven, P. H., Roberts, C. M. & Sexton, J. O. (2014), 'The biodiversity of species and their rates of extinction, distribution, and protection', *Science* **344**(6187), 1246752. doi: https://doi.org/10.1126/science.1246752.

Pimm, S. L. & Raven, P. (2000), 'Extinction by numbers', *Nature* **403**(6772), 843–845. doi: https://doi.org/10.1038/35002708.

Preston, F. W. (1962), 'The canonical distribution of commonness and rarity: Part I', *Ecology* **43**(2), 185–215. doi: https://doi.org/10.2307/1931976.

Romero, J. P. & Gramkow, C. (2021), 'Economic complexity and greenhouse gas emissions', *World Development* **139**, 105317. doi: https://doi.org/10.1016/j.worlddev.2020.105317.

Stern, D. I. (2004), 'The rise and fall of the environmental kuznets curve', *World Development* **32**, 1419–1439. doi: https://doi.org/10.1016/j.worlddev.2004.03.004.

Stojkoski, V., Koch, P. & Hidalgo, C. A. (2023), 'Multidimensional economic complexity and inclusive green growth', *Communications Earth & Environment* **4**, 130. doi: https://doi.org/10.1038/s43247-023-00770-0.

The World Bank (2020), *World Development Indicators*, The World Bank, Washington, DC.

United Nations Development Programme (2020), *Human Development Report 2020: The Next Frontier — Human Development and the Anthropocene*, United Nations Development Programme, New York. doi: https://doi.org/10.18356/9789210055161.

Vergez, A., Siikamäki, J., Mittempergher, D. & Piaggio, M. (2025), Agricultural support, biodiversity, and trade, Technical report, International Union for Conservation of Nature (IUCN). doi: https://doi.org/10.2305/rihu4909.

Vincent, J. R. (1997), 'Testing for environmental kuznets curves within a developing country', *Environment and Development Economics* **2**(4), 417–431. doi: https://doi.org/10.1017/s1355770x97000223.

Wiebe, R. A. & Wilcove, D. S. (2025), 'Global biodiversity loss from outsourced deforestation', *Nature* **639**, 389–394. doi: https://doi.org/10.1038/s41586-024-08569-5.

Wilting, H. C., Schipper, A. M., Bakkenes, M., Meijer, J. R. & Huijbregts, M. A. J. (2017), 'Quantifying biodiversity losses due to human consumption: A global-scale footprint analysis', *Environmental Science & Technology* **51**, 3298–3306. doi: https://doi.org/10.1021/acs.est.6b05296.

Online Appendix

Outsourcing Extinction: Economic Complexity and the Geography of Biodiversity Loss

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A Derivations for the theoretical framework

This appendix collects the derivations underlying the propositions, lemmas, and key equations of Section 2.

A.1 Proof of Lemma 2.1 (composition effect)

Cost minimization in region i equates the value marginal product of labor across sectors,

$$\kappa_i \, \eta \, (L_i^y)^{\eta-1} \;=\; pB\eta \, (L_i^x)^{\eta-1}, \tag{A.1}$$

which gives the labor ratio

$$\frac{L_i^y}{L_i^x} \;=\; \left(\frac{\kappa_i}{pB}\right)^{1/(1-\eta)}. \tag{A.2}$$

Imposing the labor constraint $L_i^x + L_i^y = L_i$ delivers L_i^x as in Equation (2.6). Differentiating at fixed p,

$$\frac{\partial L_i^x}{\partial \kappa_i} \;=\; -\frac{L_i}{\left[1+(\kappa_i/pB)^{1/(1-\eta)}\right]^2} \cdot \frac{(\kappa_i/pB)^{\eta/(1-\eta)}}{(1-\eta)\,pB} \;<\; 0. \tag{A.3}$$

Since $x_i = B(L_i^x)^\eta$ and $h_i = \gamma_i x_i$,

$$\frac{\partial x_i}{\partial \kappa_i} \;=\; B\eta \, (L_i^x)^{\eta-1} \frac{\partial L_i^x}{\partial \kappa_i} \;<\; 0, \qquad \frac{\partial h_i}{\partial \kappa_i} \;=\; \gamma_i \frac{\partial x_i}{\partial \kappa_i} \;<\; 0, \tag{A.4}$$

which is the statement of Lemma 2.1. ■

A.2 Derivation of the conditional effect (Eq. 2.8)

In the two-region case, the conservation status of Home is

$$b_H \;=\; \omega_{HH}\left(1-\tfrac{h_H}{\bar{A}_H}\right)^z + \omega_{HF}\left(1-\tfrac{h_F}{\bar{A}_F}\right)^z. \tag{A.5}$$

At fixed Home income I_H and prices, h_H and h_F both depend on κ_H through the trade equilibrium. Differentiating b_H with respect to κ_H,

$$\frac{db_H}{d\kappa_H} \;=\; -\,\omega_{HH}\cdot\frac{z}{\bar{A}_H}\left(1-\tfrac{h_H}{\bar{A}_H}\right)^{z-1}\frac{dh_H}{d\kappa_H} \;-\; \omega_{HF}\cdot\frac{z}{\bar{A}_F}\left(1-\tfrac{h_F}{\bar{A}_F}\right)^{z-1}\frac{dh_F}{d\kappa_H}. \tag{A.6}$$

Substituting the definition $m_j \equiv (z/\bar{A}_j)(1-h_j/\bar{A}_j)^{z-1} > 0$ from Equation (2.5) yields

$$\frac{db_H}{d\kappa_H} \;=\; -\,\omega_{HH}\,m_H\,\frac{dh_H}{d\kappa_H} \;-\; \omega_{HF}\,m_F\,\frac{dh_F}{d\kappa_H}, \tag{A.7}$$

which is Equation (2.8). The first term is non-negative, since $dh_H/d\kappa_H \leq 0$ by Lemma 2.1; the second is non-positive, since $dh_F/d\kappa_H \geq 0$ by Section A.3. Proposition I states the conditions under which the second term dominates.

A.3 World price response and the signs of $dh_H/d\kappa_H$ and $dh_F/d\kappa_H$

Let $S(p,\kappa_H,\kappa_F) \equiv x_H(p,\kappa_H)+x_F(p,\kappa_F)$ denote world supply of the habitat-intensive good, and $D(p,I_H,I_F) \equiv \beta(I_H+I_F)/p$ world demand (from Cobb–Douglas, $c_i^x = \beta I_i/p$). World market clearing,

$$S(p,\kappa_H,\kappa_F) \;=\; D(p,I_H,I_F), \tag{A.8}$$

defines the equilibrium price p implicitly. Differentiating at fixed (I_H,I_F,κ_F) and applying the implicit function theorem,

$$\frac{dp}{d\kappa_H} \;=\; -\,\frac{\partial S/\partial \kappa_H}{\partial S/\partial p - \partial D/\partial p}. \tag{A.9}$$

The denominator is the standard excess-supply slope, positive under upward-sloping supply ($\partial S/\partial p >$ 0) and downward-sloping demand ($\partial D/\partial p = -\beta(I_H+I_F)/p^2 < 0$). The numerator is negative because $\partial x_H/\partial \kappa_H < 0$ (Lemma 2.1). Hence

$$\frac{dp}{d\kappa_H} \;>\; 0, \tag{A.10}$$

the equilibrium relative price of the habitat-intensive good rises as Home grows more complex. The implied movements in habitat conversion follow from $h_i = \gamma_i x_i$:

$$\frac{dh_H}{d\kappa_H} \;=\; \gamma_H\left[\frac{\partial x_H}{\partial \kappa_H} + \frac{\partial x_H}{\partial p}\,\frac{dp}{d\kappa_H}\right] \;<\; 0, \qquad \frac{dh_F}{d\kappa_H} \;=\; \gamma_F\,\frac{\partial x_F}{\partial p}\,\frac{dp}{d\kappa_H} \;>\; 0. \tag{A.11}$$

Conversion migrates from Home to Foreign as Home grows more complex.

A.4 Derivation of Corollary 2.2 (spatial dependence)

Let $g_j \equiv (1-h_j/\bar{A}_j)^z$ denote the survival share in region j. From Equation (2.4),

$$b_i \;=\; \sum_j \omega_{ij}\,g_j. \tag{A.12}$$

Treating $\{g_j\}_{j=1}^N$ as random across countries with finite second moments, the covariance between any two regions' biodiversity indices is

$$\mathrm{Cov}(b_i,b_k) \;=\; \mathrm{Cov}\left(\sum_j \omega_{ij}\,g_j,\, \sum_\ell \omega_{k\ell}\,g_\ell\right) \;=\; \sum_{j,\ell} \omega_{ij}\,\omega_{k\ell}\,\mathrm{Cov}(g_j,g_\ell). \tag{A.13}$$

If, in addition, the $\{g_j\}$ are mutually non-negatively correlated (a property that holds when habitat conversion shocks are themselves non-negatively correlated across countries, which the data support), then every term in Equation (A.13) is non-negative, and the sum is strictly positive whenever there exists j such that $\omega_{ij}\,\omega_{kj} > 0$. This proves the first claim of Corollary 2.2.

For the spatial-autoregressive representation, let $\mathbf{b} = (b_1, \dots, b_N)^\top$ and let W be a row-standardized $N \times N$ weight matrix with $W_{ii} = 0$. The best linear projection of b_i onto the spatial lag $(W\mathbf{b})_i = \sum_{k \neq i} W_{ik}\, b_k$ is, by the standard projection theorem,

$$b_i \;=\; \rho\,(W\mathbf{b})_i + \xi_i, \qquad \rho \;=\; \frac{\mathrm{Cov}(b_i, (W\mathbf{b})_i)}{\mathrm{Var}((W\mathbf{b})_i)}, \qquad E[\xi_i\,(W\mathbf{b})_i] = 0. \tag{A.14}$$

By Equation (A.13), the numerator of ρ is non-negative whenever W inherits the structure of Ω (so that $W_{ik} > 0$ for biome-sharing pairs), and the SAR coefficient $\rho \geq 0$ measures the share of b_i's variation attributable to its spatial neighbors. The residual ξ_i is orthogonal to the spatial lag by construction.

The choice of W is empirical: in Section 4.3 we use a queen-contiguity weight matrix as a parsimonious proxy for biome-overlap; the appendix robustness checks substitute alternative weights derived from Ω directly without affecting the qualitative conclusions. ■

A.5 Proof of Proposition III (consumption-based instrument)

Planner's problem. The planner chooses $\{c_i^x, c_i^y, L_i^x, L_i^y\}_{i=1}^N$ to maximize $\mathscr{W}$ in Equation (2.12) subject to the technology $x_i = B(L_i^x)^\eta$ and $y_i = \kappa_i(L_i^y)^\eta$, the labor constraint $L_i^x + L_i^y = L_i$, the habitat-conversion identity $h_i = \gamma_i x_i$, the biodiversity index $b_i = \sum_j \omega_{ij}(1 - h_j/\bar{A}_j)^z$, and the goods-market clearing conditions. Substituting the definitional constraints into the objective yields the Lagrangian

$$\mathscr{L} \;=\; \sum_i n_i\left[\beta \log c_i^x + (1-\beta)\log c_i^y + \phi\, b_i\right] + \mu_x\left(\sum_i x_i - \sum_i c_i^x\right) + \mu_y\left(\sum_i y_i - \sum_i c_i^y\right), \tag{A.15}$$

where μ_x, μ_y are shadow world prices.

Planner's first-order conditions. Differentiating $\mathscr{L}$ with respect to L_j^x and using $\partial h_j/\partial L_j^x = \gamma_j B\, \eta\, (L_j^x)^{\eta-1}$,

$$\mu_x \cdot \frac{\partial x_j}{\partial L_j^x} \;=\; \mu_y \cdot \frac{\partial y_j}{\partial L_j^y} \;+\; \mathrm{SMC}_j \cdot \frac{\partial h_j}{\partial L_j^x}, \tag{A.16}$$

with SMC_j defined in Equation (2.13). The FOC with respect to c_ℓ^x yields the Cobb–Douglas relation $\mu_x = n_\ell\, \beta \,/\, c_\ell^x$, and similarly for c^y. The planner therefore equates the value marginal product of labor in x, net of the social marginal cost of conversion, to the value marginal product of labor in y.

Decentralized equilibrium. Firms in j choose L_j^x to maximize profit $p_x x_j - w_j L_j^x$, where w_j is the wage in region j, with FOC

$$p_x \cdot \frac{\partial x_j}{\partial L_j^x} \;=\; w_j, \tag{A.17}$$

ignoring SMC_j entirely. The wage equates across sectors, $w_j = p_y\, \partial y_j / \partial L_j^y$. A consumer in region ℓ choosing c_ℓ^x takes prices as given and perceives the private marginal cost of an extra unit as p_x, plus at most the part of conversion damage that falls on her own species, $\mathrm{PMC}_{\ell j}$ from Equation (2.14). The wedge $\Delta_{\ell j} = \mathrm{SMC}_j - \mathrm{PMC}_{\ell j} > 0$ whenever conversion in j affects species shared by countries other than ℓ.

Pigouvian instrument. Let τ_j denote a per-unit charge on the embodied habitat that consumption causes to be converted in j. The consumer's modified FOC adds τ_j to the effective marginal price of consumption originating in j. Setting

$$\tau_j = \mathrm{SMC}_j = \phi \sum_i n_i \omega_{ij} m_j \tag{A.18}$$

adds the full social marginal damage to the consumer's price, so the consumer's modified FOC coincides with Equation (A.16). The decentralized allocation under τ therefore implements the planner's first-best.

Equivalent decentralization. An international transfer from importing region ℓ to exporting region j equal to $\Delta_{\ell j}$ for each pair (ℓ, j) also implements the first-best: the transfer prices the cross-border externality at the source while leaving each region's own internalization unchanged. ■

B Validity of the Complexity Outlook Index as an instrument

To assess the validity of the Complexity Outlook Index (COI) as an instrument for economic complexity, we run a battery of tests, collected in Table B.1. First, instrument relevance is established by a strong first stage ($F = 32$), far above conventional weak-instrument thresholds. Second, and most directly bearing on the exclusion restriction, when the COI is entered into the structural equation alongside complexity its coefficient is small and statistically insignificant: the instrument has no effect on extinction risk other than through complexity. Third, a falsification test using a lead of the COI returns a null effect, so the instrument does not pick up anticipation or pre-trends in the outcome. Together these exercises support the validity of the instrument.

Table B.1: Instrument-validity tests for the Complexity Outlook Index

Test	Statistic	Implication
(1) Relevance: first-stage coef. on COI	0.381^{***} (0.068)	strong first stage, $F = 31.7$
(2) Exclusion: COI | ECI in structural eq. (3.1)	-0.002 (0.004)	no direct effect
(3) Falsification: lead of COI	-0.003 (0.003)	no anticipation/pre-trend

Notes: Six-period panel, 1993–2019, with country and period fixed effects and the controls of Equation (3.1); standard errors clustered by country. Row (1) reports the first-stage coefficient of ECI on the COI, with its significance stars, and the corresponding F-statistic. Row (2) adds the COI to the structural equation alongside ECI; an insignificant coefficient supports the exclusion restriction. Row (3) replaces the COI with its one-period lead. Coefficients carry significance stars where significant; for the exclusion and falsification rows (2)–(3), statistical insignificance is the outcome that supports instrument validity. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C Construction of the species-range overlap proxy and bridge variables

This appendix details the construction of the species-range overlap matrix $\Omega = (\omega_{ij})_{i,j}$ and the bridge variables that enter Equation (4.3) and Table 6, and reports the sensitivity of the estimates to alternative choices of Ω.

C.1 The species-range overlap proxy ω_{ij}

The main estimation uses a binary same-continent indicator,

$$\omega_{ij} = \mathbf{1}_{\{c(i)=c(j)\}}, \tag{C.1}$$

where $c(\cdot)$ assigns each country to one of the seven continents of the Natural Earth admin_0 (country-level) dataset (Africa, Antarctica, Asia, Europe, North America, Oceania, South America). The off-diagonal mean of Ω on the 161-country sample is 0.21: roughly one country pair in five shares a continent. The same-continent indicator captures, at a level the cross-section can resolve, the within-continent species sharing that arises from regional biomes contiguous with national borders and from migratory ranges linking countries within continents.

The continent indicator is one point in a family of overlap proxies built from increasingly fine biogeographic partitions:

- **Biogeographic realm.** Each country is mapped to one of the eight Wallace/Olson realms (Afrotropical, Palearctic, Nearctic, Neotropical, Indomalayan, Australasian, Oceania, Antarctic) using a fixed crosswalk on the Natural Earth SUBREGION field; ω_{ij} is the same-realm indicator. This separates, for example, Sub-Saharan Africa (Afrotropical) from Northern Africa (Palearctic), and Southern and South-Eastern Asia (Indomalayan) from Northern and Western Asia (Palearctic).

- **Biome cosine.** For each country we compute the share of its land area in each of the fourteen biomes of Olson et al. (2001), intersecting the Natural Earth boundaries with the WWF Terrestrial Ecoregions of the World shapefile. ω_{ij} is the cosine similarity of the two countries' biome-share vectors, a continuous measure in $[0,1]$.

- **Realm $\times$ biome ecoregion cells.** The same cosine construction over 63 realm-biome cells (each TEOW polygon is tagged by both a realm and a biome). This distinguishes the tropical moist forest of the Neotropical realm from the tropical moist forest of the Indomalayan realm, which share habitat type but no species.

C.2 Construction of the bridge variables

Let $M[s,d]$ denote the entry of the 2013 bilateral extinction-risk footprint matrix (Irwin et al. 2022), giving the extinction risk borne by source s's species on account of consumption in destination d. For each destination i we construct the three bridge variables that enter Table 6:

$$F_i^{\text{imp}} = \sum_{s \neq i} M[s,i] \qquad \text{(imported footprint: own consumption falling abroad),} \tag{C.2}$$

$$F_i^{\omega} = \sum_{s \neq i} \omega_{is} M[s,i] \qquad \text{(overlap-weighted imported footprint),} \tag{C.3}$$

$$\sigma_i = F_i^{\omega} \,/\, F_i^{\text{imp}} \qquad \text{(structural overlap share of imports).} \tag{C.4}$$

On the 161-country sample, F_i^{imp} averages roughly 4,450 extinction-equivalents with a median of 1,005, and σ_i averages 0.04 with a median of 0.02. The dependent variable in Table 6 is the long-difference $\Delta \log RLI_i$ between the 1993–95 and 2018–19 windows of the six-period panel.

C.3 Sensitivity to the overlap proxy

Tables C.2–C.4 re-estimate the three columns of Table 6 using each of the alternative overlap proxies described in Appendix C.1. The pattern is consistent across proxies. The direct effect of the imported footprint (column 1) is identical to the main estimate at -0.011 ($p < 0.01$), as expected: this column does not depend on ω. The overlap-weighted imported footprint (column 2) carries the predicted negative sign in every specification, with point estimates close to the main estimate of -0.010: -0.010 for the realm proxy, -0.010 for the biome-cosine proxy, and -0.006 for the realm-by-biome cell proxy. All three are statistically significant at the one-percent level.

The overlap share (column 3) is the most sensitive to the choice of ω. The point estimate is consistently negative across all proxies (-0.080 for realm, -0.160 for biome cosine, -0.111 for realm-by-biome cell, against -0.153 for the continent proxy in Table 6), but statistical significance is achieved only for the coarser continent proxy. Finer biogeographic partitions deliver fewer source countries per bin and reduce within-importer variation in the structural share, producing wider standard errors. The model-predicted sign is preserved throughout, and the column-1 and column-2 results confirm that the bridge mechanism is robust to the choice of overlap proxy.

Table C.2: Bridge sensitivity to the overlap proxy: biogeographic realm

	(1) Direct	(2) ω-weighted	(3) Overlap share
log Imported footprint (F^{imp})	-0.011***		-0.011***
	(0.002)		(0.002)
log ω-weighted imported (F^{ω})		-0.010***	
		(0.002)	
Overlap share of imports (σ)			-0.080
			(0.071)
Δ ECI	-0.016***	-0.017***	-0.016***
	(0.005)	(0.006)	(0.005)
Δ log GDP per capita	-0.040***	-0.042***	-0.040***
	(0.011)	(0.011)	(0.011)
Δ log Population density	-0.050***	-0.057***	-0.051***
	(0.012)	(0.012)	(0.012)
Δ HDI	0.212**	0.212**	0.212**
	(0.100)	(0.107)	(0.100)
R^2	0.336	0.300	0.341
Observations	118	117	118

Notes: Replicates the three columns of Table 6 using the same-realm indicator over the 8 Wallace/Olson biogeographic realms as ω_{ij}. The realm assignment is the dominant realm in the TEOW area overlay. Each country's overlap-weighted imported footprint counts only imports from countries in the same realm. Heteroskedasticity-robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.3: Bridge sensitivity to the overlap proxy: biome cosine

	(1) Direct	(2) ω-weighted	(3) Overlap share
log Imported footprint (F^{imp})	-0.011***		-0.011***
	(0.002)		(0.002)
log ω-weighted imported (F^{ω})		-0.010***	
		(0.002)	
Overlap share of imports (σ)			-0.160
			(0.115)
Δ ECI	-0.016***	-0.015***	-0.015***
	(0.005)	(0.006)	(0.005)
Δ log GDP per capita	-0.040***	-0.041***	-0.040***
	(0.011)	(0.011)	(0.011)
Δ log Population density	-0.050***	-0.046***	-0.047***
	(0.012)	(0.012)	(0.012)
Δ HDI	0.212**	0.218**	0.207**
	(0.100)	(0.106)	(0.100)
R^2	0.336	0.325	0.349
Observations	118	118	118

Notes: Replicates the three columns of Table 6 using the cosine similarity of countries' 14-biome share vectors as ω_{ij}. Biome shares are computed from the WWF Terrestrial Ecoregions of the World, collapsed across realms. Heteroskedasticity-robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.4: Bridge sensitivity to the overlap proxy: realm-by-biome cells

	(1) Direct	(2) ω-weighted	(3) Overlap share
log Imported footprint (F^{imp})	-0.011***		-0.011***
	(0.002)		(0.002)
log ω-weighted imported (F^{ω})		-0.006***	
		(0.002)	
Overlap share of imports (σ)			-0.111
			(0.112)
Δ ECI	-0.016***	-0.018***	-0.015***
	(0.005)	(0.007)	(0.005)
Δ log GDP per capita	-0.040***	-0.046***	-0.041***
	(0.011)	(0.012)	(0.011)
Δ log Population density	-0.050***	-0.053***	-0.050***
	(0.012)	(0.013)	(0.012)
Δ HDI	0.212**	0.300***	0.216**
	(0.100)	(0.112)	(0.099)
R^2	0.336	0.247	0.341
Observations	118	118	118

Notes: Replicates the three columns of Table 6 using the cosine similarity of countries' shares over 63 realm-by-biome ecoregion cells as ω_{ij}. This is the most biologically detailed proxy considered: tropical moist forest in the Neotropical realm and tropical moist forest in the Indomalayan realm are treated as distinct cells because they share habitat type but no species. Heteroskedasticity-robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D Sample, functional form, and frequency

This appendix reports the sample, functional-form, and frequency checks that support the robustness exercises of Section 4.3. The sequential additions of controls, first-link tests for domestic channels, and spatial-dependence checks are reported in the main text. Table D.5 reports the full coefficient vector across four variants of the preferred specification: the baseline six-period panel (column 1), the balanced subsample of countries observed in all six periods (column 2), the EKC test that adds a quadratic in log GDP per capita (column 3), and the three-period aggregation (column 4). The Economic Complexity Index coefficient is stable across all four columns. In the EKC column, both the linear and the quadratic income terms are statistically insignificant, so the data do not support an inverted-U in income.

Table D.5: Further robustness checks

	(1) Preferred	(2) Balanced panel	(3) EKC quadratic	(4) Three-period panel
Economic Complexity Index	-0.012***	-0.013***	-0.012***	-0.015***
	(0.004)	(0.004)	(0.004)	(0.005)
log GDP per capita	-0.020**	-0.033***	-0.059	-0.033***
	(0.008)	(0.009)	(0.044)	(0.009)
log GDP per capita squared			0.002	
			(0.002)	
log Population density	-0.030***	-0.044***	-0.025**	-0.041***
	(0.011)	(0.012)	(0.012)	(0.011)
Human Development Index	0.134**	0.179**	0.151**	0.189***
	(0.061)	(0.077)	(0.060)	(0.066)
Within R^2	0.418	0.446	0.420	0.445
Observations	1,000	774	1,000	484

Notes: Columns report the within-country specification Equation (3.1) with country and period fixed effects. Column 1 is the preferred six-period panel. Column 2 restricts the sample to countries observed in all six periods. Column 3 adds log GDP per capita squared to test the biodiversity Environmental Kuznets Curve; neither the linear nor the quadratic income term is statistically significant. Column 4 re-estimates the model on three longer periods (1993–95 / 2005–07 / 2017–19) instead of the preferred six-period panel. Standard errors clustered by country in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.